After Berlin's Electoral Breakdown: Voting and Precinct Organization*

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Abstract

How does voting change after an electoral breakdown? Using court-documented exposure and Berlin election returns from 2014 to 2026, I compare turnout in court-linked and clean-control precincts. Court-linked areas exhibit persistently lower in-person turnout and higher postal turnout in the 2024 European and 2025 federal elections. Between the 2017 and 2025 federal elections, treated–control differences change by −1.77 percentage points in person and +1.25 by post; the total-turnout estimate is smaller and imprecise (−0.52 points). Pre-breakdown changes between the 2014 and 2019 European elections were substantially smaller. Complementary geographic evidence shows a steeper association between baseline precinct electorate size and subsequent catchment reorganization in court-linked areas. The findings distinguish persistent voting-mode differences from changes in electoral administration.

Keywords: election administration; institutional adjustment; turnout; voting costs; postal voting; electoral failure; difference-in-differences

JEL classifications: D72; D73

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1 Introduction

Electoral participation depends not only on citizens' incentives but also on the administration that makes voting possible. Missing ballots, congested polling places, and interrupted voting can raise the cost of participation on election day. But what happens after these failures? Citizens may continue voting in person, switch to postal voting, or abstain, while electoral administrators may redraw polling catchments, reassign residential areas, or revise polling operations. These responses may occur together without necessarily restoring the accessibility or reliability of in-person voting. I study persistent differences in local voting-mode composition after a documented breakdown and, as a complementary institutional outcome, subsequent changes in polling-catchment geography.

When voters can cast their ballots either in person at a polling station or by post, it is important to distinguish between *participation* and *voting mode*. Higher expected polling-place costs need not reduce total turnout one-for-one if postal voting is a viable alternative, although using that channel can require information and advance planning. On the institutional side, redrawing polling catchments changes which residential locations belong to the same in-person voting precinct, but need not increase voting capacity or improve access. Reorganization may also be more extensive where precincts initially serve larger electorates. Behavioral adjustment, administrative reconfiguration, and effective remediation are therefore distinct outcomes.

Berlin's September 2021 elections provide an unusually informative setting. Concurrent federal, state, district, and referendum votes were affected by queues, interruptions, closures, and missing or incorrect ballots. The combination of elections with different stakes is itself relevant to electoral participation (Cantoni, Gazzè, & Schafer, 2021). The Berlin Constitutional Court and the German Federal Constitutional Court subsequently documented failures at geographically identifiable locations (Bundesverfassungsgericht, 2023; Verfassungsgerichtshof des Landes Berlin, 2022). I combine these records with official election returns from 2014 through 2026, harmonized to the geography of 2021 postal precincts. The participation analysis compares precincts linked to the federal court record with precincts not identified in either court record, using precinct and election-by-district fixed effects. It follows the 2021 breakdown through the 2023 state rerun and later regular elections, including the 2024 European election but excluding the February 2024 partial federal rerun from the main panel. The estimates describe changes in local turnout trajectories associated with court-documented exposure.

The principal finding concerns the composition of electoral participation. Between the 2014 and 2019 European elections, treated–control turnout differences changed relatively

little (Appendix B). Relative to the 2019 European-election benchmark, court-linked areas exhibit persistently more negative changes in in-person turnout than comparison areas, alongside relatively higher postal turnout in the 2024 European and 2025 federal elections. The comparison within federal elections is especially informative: from 2017 to 2025, the treated–control difference changes by −1.77 percentage points for in-person voting and +1.25 points for postal voting. Their sum is a smaller, less precise total-turnout estimate of −0.52 percentage points. Across elections, the clearest pattern is a persistent difference in local voting-mode composition.

A complementary finding concerns the reorganization of in-person polling catchments. Using a fixed grid of residential locations, I compare the population-weighted composition of the polling catchment serving each location in 2021 with the catchment serving that same location in later elections. A higher Jaccard-loss value indicates less overlap between the two catchments. By 2026, the relationship between original precinct electorate size and subsequent reorganization is steeper in court-linked areas: the preferred conditional interaction is +7.20 percentage points of catchment loss per standard deviation of baseline electorate size. This pattern reflects especially limited reorganization among initially smaller court-linked precincts, while initially larger precincts exhibit similar levels of reorganization across exposure groups. A separate analysis tracks the electorate assigned to polling precincts, a proxy for potential administrative workload.

Several checks assess the sensitivity of these findings. For participation, the 2014 and 2019 European elections provide a same-election-family benchmark before the breakdown, alongside the 2016 state and 2017 federal elections. Socioeconomic controls assess sensitivity to observed compositional differences. Within-election-family comparisons and alternative control samples assess electoral comparability and exposure classification. I also reproduce the in-person results on original polling-station geography to probe sensitivity to spatial harmonization. For administrative reorganization, nonlinear controls for initial precinct size and alternative measures of catchment overlap assess whether the workload interaction depends on the construction of the baseline geography. District-level inference complements the main precinct-clustered estimates.

These findings contribute to research on voting costs and electoral institutions.[1] The standard voting calculus emphasizes the private costs of participation (Downs, 1957; Riker & Ordeshook, 1968). Empirical studies show how registration requirements, travel distances, and polling-place conditions shape turnout (Braconnier, Dormagen, & Pons, 2017; Brady & McNulty, 2011; Cantoni, 2020; Clinton, Eubank, Fresh, & Shepherd, 2021). Alternative vot-

[1]For an overview of voting procedures, electoral rules, and participation, see Cantoni, Pons, and Schafer (2025).

ing channels introduce another margin: citizens can change how they participate rather than simply whether they participate (Bagwe, Margitic, & Stashko, 2026; Barber & Holbein, 2020; Gerber, Huber, & Hill, 2013). Evidence on internet voting in Estonia similarly highlights the distinction between the costs of in-person voting and the use of an alternative channel (Solvak, Vassil, Vinkel, & Romanov, 2026).

A related literature examines the persistence of voting behavior and the consequences of prior electoral experiences. Evidence on habit formation and long polling-place waits suggests that earlier experiences can influence subsequent participation (Fujiwara, Meng, & Vogl, 2016; Gerber, Green, & Shachar, 2003; Pettigrew, 2021). Especially close to this study, Alipour and Lindlacher (2025) examine persistent voting-mode changes following planned polling-place reassignments, while Tomkins et al. (2023) study the effects of polling-place assignment. Fischer and Flügel (2025) analyze the immediate turnout consequences of Berlin's 2021 breakdown. This paper addresses a distinct question: whether local voting-mode differences persist across subsequent elections after an unexpected administrative failure and its legal aftermath. Unlike planned polling-place reassignment, the Berlin episode involved disruptions to the conduct of an election followed by judicial review; unlike the existing study of the Berlin episode, the analysis follows participation in later elections. The complementary catchment analysis documents how the geography of in-person voting evolved over the same period.

The administrative findings further connect the paper to research on the organization of electoral institutions. Institutional restructuring can affect participation, as illustrated by evidence on municipal mergers in France (Mille, 2026), although changing political jurisdictions differs from redrawing polling precincts. Existing work primarily studies how electoral arrangements shape participation. Considering both dimensions distinguishes changes in the use of voting opportunities from changes in their administrative provision.

The paper proceeds as follows. Section 2 develops voting-mode choice and administrative adjustment as distinct decisions. Section 3 describes the institutional episode. Section 4 explains the geographic data and empirical designs. Section 5 presents voting-mode outcomes and identification diagnostics. Section 6 examines catchment reorganization and electorate workloads. Section 7 concludes.

2 Voting costs, voting modes, and administration

The standard voting calculus compares instrumental and civic benefits with the private cost of casting a ballot (Downs, 1957; Riker & Ordeshook, 1968). Administrative failures

raise realized time costs and uncertainty for some citizens. With postal voting available, participation becomes a choice among three alternatives: in-person voting, postal voting, and abstention. Let $m \in \{IP, P\}$ denote in-person and postal voting, respectively, with the outside option of abstaining normalized to zero. The return to each voting mode is

$$R_{imt} = p_{it} B_{it} + D_{it} - C_{imt}. \tag{1}$$

Here, C_{imt} denotes the anticipated private cost of using voting mode m at the time of the participation decision. A citizen participates when at least one mode provides a positive return, choosing the higher-return mode when both do:

$$\text{Vote}_{it} = \mathbf{1}\left\{ \max_{m \in \{IP,P\}} R_{imt} > 0 \right\}, \qquad m^*_{it} = \arg \max_{m \in \{IP,P\}} R_{imt} \quad \text{if Vote}_{it} = 1. \tag{2}$$

The common-benefit specification is deliberately parsimonious. Ballot validity, the expressive benefits of voting at a polling place, or confidence in a specific voting method could also vary by mode. The essential distinction is that a higher relative cost of in-person voting need not reduce participation if postal voting remains an acceptable and sufficiently inexpensive substitute.

An adverse experience can affect the expected cost of later in-person elections, even if objective polling-place conditions subsequently improve. Write

$$\widehat{C}^{IP}_{i,t+k} = C^{IP,0}_{i,t+k} + \rho_k F_i, \qquad k \geq 1, \tag{3}$$

where $C^{IP,0}_{i,t+k}$ denotes the counterfactual expected in-person voting cost in the absence of the failure-related adjustment, F_i denotes exposure to the failure, and ρ_k the possible additional expected in-person cost at horizon k. The sign and persistence of ρ_k depend on beliefs, experience, and subsequent information; $\rho_k = 0$ corresponds to the disappearance of the failure-related increment in expected in-person voting costs. The expression is an interpretive device, not an estimated updating equation. Individual perceptions, the initial experience of each voter, and the adjustment parameter are not observed in the precinct returns.

Two distinctions matter for the empirical outcomes. First, a relative increase in expected in-person cost predicts a lower incentive to use the polling place, while its effect on total participation depends on whether postal voting remains preferable to abstention. Second, the availability of postal voting does not make adjustment costless. Applications, deadlines, and advance planning can entail procedural costs (Menger & Stein, 2020; Stewart, 2010). Familiarity with the process may affect the speed at which voters adopt it, giving a possible

reason for postal compensation to emerge only in later elections. This is a hypothesis about choice costs; the aggregate data do not establish that particular voters learned or switched modes.

The contrast with a general loss of confidence in elections is also relevant. A shock that raises perceived costs or reduces expected benefits across *both* modes would predict weaker compensation through postal voting than a shock specific to polling-place access. Research on election administration and voter confidence provides reasons to take this alternative seriously (Alvarez, Cao, & Li, 2021; Atkeson & Saunders, 2007; Bowler, Brunell, Donovan, & Gronke, 2015), but the present design cannot separately estimate a general confidence channel.

The framework distinguishes three observable outcomes. Persistent negative in-person estimates can coexist with positive postal estimates and little change in total turnout; alternatively, postal compensation can be insufficient, leaving negative total-turnout differences. If voting costs and other determinants of participation returned to their counterfactual paths, the model would predict no persistent mode-specific differences attributable to the failure. The relative responses of access-cost and ballot-error exposure provide an additional, more tentative comparison because their administrative classification and affected samples differ. These are implications for local participation aggregates, not identified individual transitions.

The supply of in-person voting is also organized administratively. Electoral officials assign residential locations to polling precincts and can reorganize catchments through reassignment, splitting, or consolidation. These decisions may reflect registered electorates, geographical constraints, available buildings and personnel, demographic change, and the costs of reorganization. Larger initial electorates may create greater pressure or opportunities for reorganization. Following a documented breakdown, administrators may also have reasons to review previously disrupted precincts, although their motivations are not observed. This motivates examining whether the relationship between baseline electorate size and subsequent catchment reorganization differs between court-linked and clean comparison areas. Such a difference would not, by itself, establish intentional reform targeting.

Let $a_t(g)$ denote the original in-person polling precinct assigned to residential grid cell g in election t, and let $W_t(a_t(g))$ denote the electorate registered in that precinct. Changes in precinct identifiers alone need not represent substantive reorganization; the empirical analysis therefore measures changes in the population-weighted overlap of the catchments serving fixed residential locations. Changes in $W_t(a_t(g))$, by contrast, measure the size of the assigned electorate, which serves as a proxy for administrative workload. Neither outcome directly captures staffing, polling-place capacity, queue lengths, travel distance, or individual

voting costs. Moreover, changes in assigned electorates can reflect demographic developments or franchise rules rather than administrative decisions. Electoral reorganization and voting-mode adjustment are thus distinct outcomes of the post-breakdown period, neither of which necessarily establishes effective remediation.

3 Berlin's 2021 Electoral Breakdown

On September 26, 2021, Berlin held federal, state, district, and referendum votes on the same day, alongside the Berlin Marathon and under pandemic-related restrictions. Election officials had to distribute multiple ballots and organize voting across many local polling locations. The resulting administrative breakdown was not a uniform citywide closure but a geographically uneven set of problems. Documented incidents included missing ballots, incorrect ballot materials, queues, temporary interruptions and closures, and voting after the statutory closing time (Bundesverfassungsgericht, 2023; Verfassungsgerichtshof des Landes Berlin, 2022). They differ substantively: queues and interruptions can impose immediate time and uncertainty costs; missing ballots can delay or prevent the act of voting; incorrect ballot materials implicate the validity or correctness of the vote even when physical access is possible.

The breakdown generated two court proceedings that form the empirical record. The Berlin Constitutional Court annulled the state election, which was rerun in 2023. The German Federal Constitutional Court subsequently ordered a partial federal rerun, held in 2024, following its review of the federal election (Bundesverfassungsgericht, 2023; Verfassungsgerichtshof des Landes Berlin, 2022). The 2024 partial rerun is not included as a regular citywide election in the event-study panel because it did not provide an equivalent vote for the same set of affected and unaffected precincts. Subsequent regular federal, European, and state elections instead provide common comparisons across the original 2021 treatment geography. The 2023 state rerun and later regular elections are analytically useful follow-ups, not separately identified effects of the legal remedies.

The court materials serve distinct measurement purposes. The federal court-linked record determines the preferred binary exposure definition. The state record is more informative about the type and reported severity of polling-place disruption. The two sources overlap but are not identical: some areas appear in both records, some appear in the federal record alone, and others appear only in the state record.[2] The primary specification excludes state-only

[2]Appendix A provides additional institutional background and documents the overlap between the court records, the disruption classification, and the construction of the electoral geography.

units from its control group. This avoids treating areas with state-documented problems as clean controls simply because they are absent from the federal record. State-record families and waiting/closure measures are used in supporting analyses with correspondingly different samples rather than as interchangeable definitions of the primary treatment.

Postal voting is an established participation channel in German elections. It can avoid the need to attend a polling place on election day but normally entails requesting and returning ballot documents in advance. The timing matters for causal interpretation. Many postal votes in September 2021 were cast before the failures encountered at in-person locations became observable, so a negative 2021 postal coefficient in a treatment subgroup cannot itself represent substitution in response to an election-day queue or closure. Post-2021 elections provide the more relevant evidence on adjustment across modes. This does not rule out general pre-election concerns or contemporaneous administrative problems affecting other channels.

The episode was also unusually visible. Berlin respondents in the German Longitudinal Election Study reported concerns about electoral fairness and correctness and elevated reports of having been prevented from voting (German Longitudinal Election Study (GLES), 2023). This is contextual evidence that the administrative failure was salient. Because the survey cannot be matched to the court-linked precinct exposure used here, it does not reveal whether personally affected voters subsequently changed their expectations or preferred voting mode.

The institutional setting therefore provides three elements for an empirical analysis: documented geographic variation in administrative problems, repeated official turnout outcomes by voting mode, and later elections after the original failure and its judicial review. It also creates an important limitation. Court-linked exposure refers to a location associated with a documented problem, not to an individually verified experience of electoral disruption. Precinct-level estimates may incorporate changes in local electorates, subsequent administrative decisions, and the public response to the failure.

4 Methodology

4.1 Data and Treatment Construction

The analysis combines official Berlin election returns for ten elections from 2014 through 2026 with the federal and state court-linked records of the 2021 breakdown. The returns are harmonized to 2021 postal-precinct geography (BWB21). The preferred comparison

comprises 294 federal-court-linked precincts and 1,056 comparison precincts not identified in either court record, for a balanced total of 13,500 precinct-election observations across 1,350 units. All three turnout outcomes have complete observations in the preferred 2014–2026 sample. The complete panel-construction accounting, source-overlap counts, and summary statistics are reported in Appendix A.

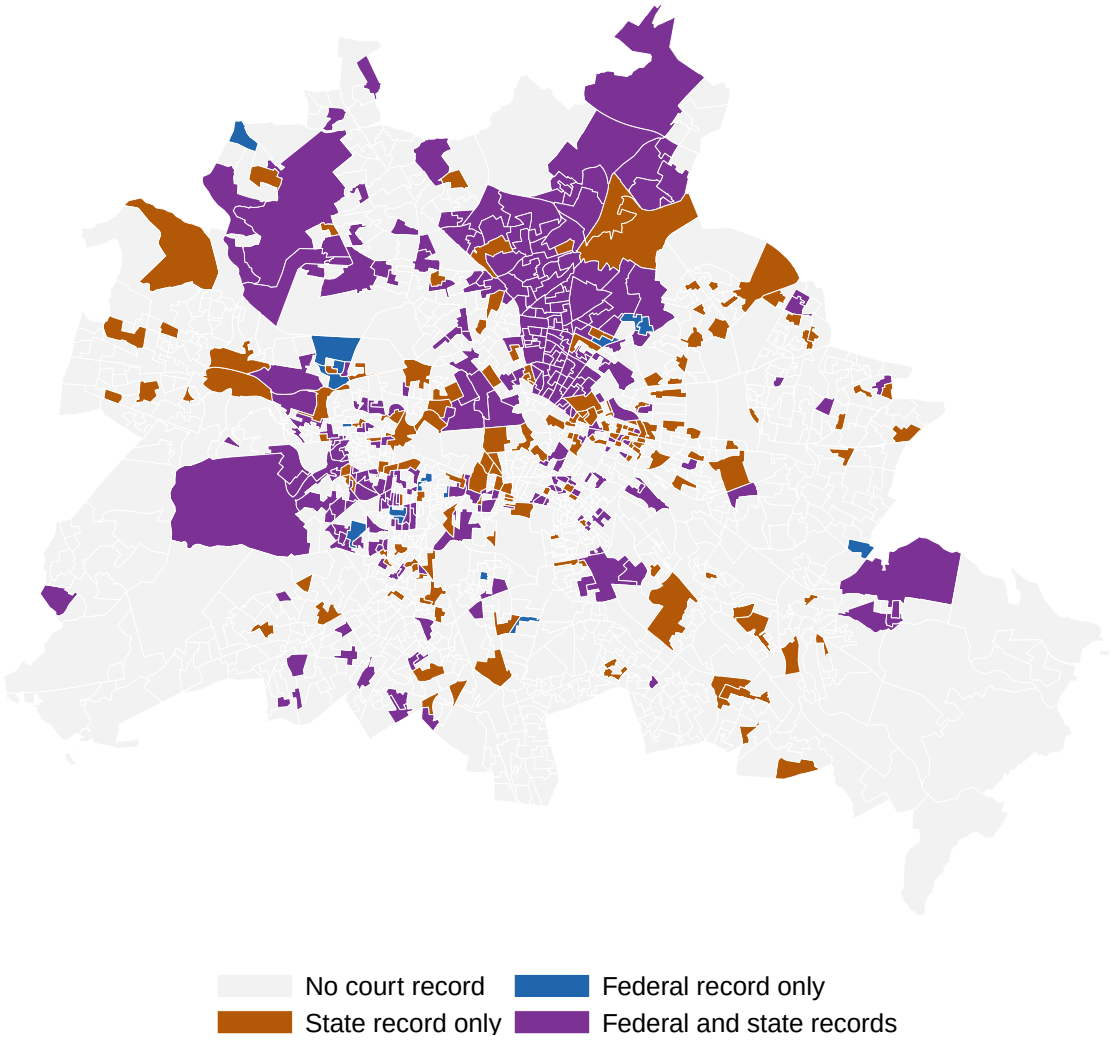
Two geographical reporting levels must be distinguished. Election-day ballots are recorded by polling-station precinct (*Urnenwahlbezirk*, UWB), whereas postal votes are reported by postal precinct (*Briefwahlbezirk*, BWB). Their assignment and spatial boundaries are not constant across elections. The target units for the primary panel are the 2021 BWB polygons.

The preferred binary treatment follows the federal court-linked 2021 polling-station precinct assignments. The exact UWB-to-BWB21 mapping and exposure-share thresholds are documented in Appendix A. Clean controls are absent from both court-record sources, while precincts documented only in the state record are excluded. This definition is conservative about contamination in the comparison group, at the cost of excluding some areas and conditioning on administrative documentation. The corresponding geographic categories appear in Figure 1. They should not be interpreted as a map of voters who personally experienced an interruption or as random assignment of administrative failures.

An election-specific source unit's in-person voter counts and official registered-electorate counts are allocated from UWB reporting units to intersecting BWB21 target precincts using population-weighted crosswalks. Postal voter counts are allocated separately from their original BWB reporting units to the same target geography. For each target precinct and election, in-person, postal, and total turnout are then calculated using *the same* harmonized official registered-electorate denominator, derived from the UWB-level electorate counts. Total turnout equals the sum of in-person and postal turnout, so their estimated coefficients also add up in identical regression samples. The spatial weights are based on a 2022 grid of German citizens aged 18 and older. They allocate *where* election-specific administrative counts are assigned, but do not replace the official registered electorate as the turnout denominator. In particular, weights summing to one ensure that citywide counts are conserved without proving that the distribution across target precincts is measured without spatial interpolation error.

The extent of interpolation varies across elections. The 2026 source boundaries generate more target intersections than the relatively concentrated 2024–2025 crosswalks, although earlier election geographies also involve substantial many-to-many allocation. The measurement audit records source-weight sums, overlap counts, and 18 harmonized precinct-election observations with total turnout above 100 percent (Appendix Table A4; Appendix Table A7).

Figure 1: Court-linked treatment exposure in Berlin



Notes: The figure maps 2021 postal precincts by the four observed combinations of federal and state court-record coverage: neither record (1,056 areas), federal record only (17), state record only (157), and both records (277). The 294 federally documented areas enter the preferred treated group; the 157 state-only areas are excluded from that comparison, not shown as a separate fifth "ambiguous" category. Source-overlap counts are reported in Appendix Table A1.

The direct 2014 source reconciliation and treated/control interpolation-concentration comparisons are reported in Appendix Table A5 and Appendix Table A6. Concentration-weighted and restricted-crosswalk checks are examined with the results. These are reasons to validate the geographic approximation rather than reasons to assume every allocated precinct rate is an exact observed administrative quantity.

Electoral eligibility introduces a separate comparability issue. In the 2024 European Parliament election, the voting-age threshold was 16 and eligible non-German EU citizens could vote; Berlin's 2026 state election also extended eligibility to German citizens aged 16 and older.[3] Official denominators reflect each election's rules, but the fixed geographic population grid is not perfectly matched to these electorates. The resulting concern has two components: the participation behavior of the eligible population may change when the franchise expands, and the spatial distribution of newly eligible voters may differ from the grid used for allocation. A fully age-matched alternative crosswalk requires verified high-resolution counts and is not reported as completed in their absence. Within-federal and within-state comparisons help characterize election-family sensitivity but do not by themselves remove the 2026 franchise change.

State records additionally distinguish waiting, voting after closing, interruptions, missing ballots, and ballot errors. The access-cost classification groups events plausibly impeding the ability to vote in person, while the ballot-error-only category contains documented errors without a recorded access-cost incident. A separate waiting/closure measure uses the maximum documented duration within an original polling-station precinct and aggregates it to BWB21 with eligible-voter weights. Federal-only observations lacking state-record severity data are not assigned zero severity. These analyses therefore have distinct measurement assumptions and smaller or differently composed treatment samples than the main court-linked specification.

4.2 Empirical Strategy

The preferred specification compares the evolution of turnout in court-linked affected and clean-control precincts while allowing districts to experience different election-specific shocks. For outcome Y_{it} in 2021 postal precinct i and election t, I estimate

$$Y_{it} = \alpha_i + \lambda_{d(i)t} + \sum_{e \neq 2019} \beta_e \left(\text{Treated}_i \times \mathbf{1}\{t = e\} \right) + \varepsilon_{it}. \tag{4}$$

[3]For the 2024 European election, see https://www.bundeswahlleiterin.de/europawahlen/2024/informationen-waehler.html; for the 2026 Berlin election, see https://www.berlin.de/wahlen/wahlen/berliner-wahlen-2026/fragen-und-antwortkatalog/artikel.1646712.php.

The three dependent variables are in-person, postal, and total turnout measured as percentage points of registered voters. Precinct effects α_i absorb stable local differences, including persistent differences in average political participation. Election-by-district effects $\lambda_{d(i)t}$ absorb common changes within each Berlin district at each election. Coefficients β_e measure changes in the treated–control turnout difference relative to the 2019 European election, conditional on these fixed effects. The main standard errors are clustered by 2021 postal precinct. Full baseline coefficients appear in Appendix Table B1; the implications of district-level inference are assessed below.

The estimand is a location-level response to documented exposure, not the average effect of an individually verified queue or ballot shortage. The analysis fixes treatment according to the 2021 court records, while later electorates may contain different individuals. Accordingly, differences in mode-specific turnout describe aggregate changes in the use of voting channels, not observed individual transitions between voting in person, voting by post, and abstaining. The treatment also includes potential reactions to publicity and legal proceedings following the breakdown. Neither a specific expectation-updating parameter nor the effect of the judicial remedies can be separately recovered from this design.

The identifying assumption is that, absent the breakdown and its associated aftermath, treated and clean-control precincts would have followed comparable turnout changes within the same district and election. This is demanding because administrative failures and their documentation need not be random. Unadjusted treated–control turnout gaps are displayed in Appendix Figure B1. The 2019 balance statistics show that exposed precincts are larger on average (about 1,884 residents versus 1,642), with differences in age and socioeconomic composition (Appendix Table B2). Baseline imbalance is not itself a violation of parallel trends; the concern is whether characteristics associated with court-documented exposure also predict different changes in turnout absent the breakdown, particularly in postal voting. Socioeconomic adjustments, matching, and comparison-group restrictions probe observable sources of such differences, but cannot rule out divergent trends arising from unobserved characteristics.

The three non-reference pre-treatment elections—2014 European, 2016 AGH, and 2017 federal—provide checks on differential trajectories. The 2014–2019 European comparison is especially informative because it holds election type constant: its treated–control changes are -0.33 points for in-person turnout, $+0.25$ for postal turnout, and -0.07 for total turnout, each considerably smaller in magnitude than the corresponding 2019–2024 European changes. The 2016 total-turnout coefficient is less reassuring. These diagnostics complement the conditional parallel-trends assumption; a same-family historical comparison remains informa-

tive rather than decisive. Pre-treatment estimates, within-district reassignment diagnostics, descriptive magnitude calibration, and district-clustered uncertainty are reported in Appendix B.

Several additional analyses distinguish robustness of the main estimates from robustness of their interpretation. Within-election-type comparisons assess sensitivity to using the common 2019 European-election reference across different electoral contexts; more concentrated crosswalk samples and original UWB geography test sensitivity to spatial allocation; alternative clustering and comparison samples address uncertainty and possible local contamination (Appendix B). Finally, the separate state-record models compare disruption types and measured waiting/closure severity. Those specifications are supplementary and are not alternative estimates of exactly the same exposure contrast.

4.3 Administrative reorganization design

The administrative analysis is distinct from the turnout event study. I use a fixed 100-meter residential grid, assigning every populated cell g to its original in-person precinct (UWB) in each observed election. Let $C_t(g)$ be the set of residential cells belonging to the UWB serving g in election t, and let $P(S) = \sum_{h \in S} p_h$ be the total population in cell set S, where p_h is the same fixed 2022 count of German citizens aged 18 or older throughout the analysis. For comparisons with the 2021 state election, the population-weighted Jaccard loss is

$$L_{gt} = 1 - \frac{P(C_{2021}(g) \cap C_t(g))}{P(C_{2021}(g) \cup C_t(g))}. \tag{5}$$

The numerator sums population in cells belonging to both original catchments; the denominator sums population in their union. The measure lies between zero (identical catchments) and one (no overlap). It is assigned to each populated residential cell before aggregation, not calculated from renamed UWB identifiers or from polygon area alone. A population-weighted mean across cells therefore describes the exposure of the fixed residential population to changes in its assigned catchment, not the proportion of independently redrawn administrative precincts. In 2021, $L_{g,2021} = 0$ by construction; the 2023 repeat inherits 2021 geography and is not a new boundary observation. Appendix G reports the 2026 distribution and group means to place the regression magnitudes in context.

The preferred joint specification is

$$L_{gt} = \alpha_g + \lambda_{d(g)t} + \sum_{e \neq 2021} \tau_e D_g \mathbf{1}\{t = e\} + \sum_{k \in \mathcal{K}} \sum_{e \neq 2021} \Big[\gamma_{ke} z_{kg} + \delta_{ke}(D_g z_{kg})\Big] \mathbf{1}\{t = e\} + u_{gt}, \tag{6}$$

where D_g denotes the court-linked exposure of the fixed 2021 BWB21 area containing cell g. The baseline measures z_{kg} in $\mathcal{K}$ comprise the 2019 original-UWB electorate, 2019 actual voters, 2021 original-UWB electorate, and fixed-grid population of the 2021 original-UWB catchment. Each is transformed using $\log(1 + x)$ and then standardized using population-weighted means and standard deviations for the 2021 treated-and-control residential cells. These baseline measures are held constant across elections. The model includes grid-cell fixed effects α_g, district-by-election effects $\lambda_{d(g)t}$, every lower-order election interaction, and the exposure-by-baseline interactions. It weights cells by p_g. The coefficient $\delta_{2021\,\text{electorate},2026}$ measures the additional 2026 slope of catchment loss with respect to standardized 2021 precinct electorate size in court-linked relative to comparison areas, conditional on the other interacted baseline measures. It is *not* the average treated–control difference in reorganization. Since the outcome is measured relative to 2021 catchments and is zero by construction in the reference year, the 2026 interaction describes a conditional difference in reorganization gradients, not a causal difference-in-differences effect of the breakdown on administrative decisions. The main inference clusters by the original BWB21 area; district-clustered uncertainty and wild-cluster bootstrap sensitivity are reported in Appendix G. The administrative estimation uses approximately 36,600 classified residential cells, nested within 1,350 BWB21 areas and 12 districts.

Because 2021 electorate size and 2026 change share a baseline catchment, the interaction can partly reflect original-size composition, the mechanics of overlap normalization, or more general administrative reconfiguration. In addition to the existing alternative-baseline and catchment-population controls, Appendix G examines nonlinear baseline-size terms and an alternative measure that normalizes overlap by the *original* catchment population alone. These are targeted functional-form and measurement diagnostics, not instruments or an identification strategy for targeted reform. An additional interpretive issue arises for comparisons dated before 2021: because every catchment is measured against the *2021* reference geography, earlier-year differences are retrospective distance from that reference. They are not pre-treatment event-study coefficients for the administrative response, and they cannot establish that the 2021 failure initiated later boundary changes.

A complementary outcome measures the number of eligible voters assigned to the election-

year UWB serving each fixed residential grid cell in 2021 and 2026. These values are aggregated to the 1,350 BWB21 areas using fixed 2022 population weights, and their log change is regressed on disruption, standardized baseline UWB electorate, and their interaction. Adjusted and joint models condition on earlier electorate/voter measures and baseline catchment population, with district fixed effects; BWB21- and district-clustered uncertainty and wild bootstrap checks are reported in Appendix G. The 2026 state election includes newly eligible German citizens aged 16–17, whereas the 2021 electorate did not. Thus changes in this proxy combine reassignment and changes in the eligible population; neither electorate counts nor Jaccard loss directly measure booths, staffing, queuing, or effective service capacity. The two administrative outcomes are compared descriptively, not as an identified mediation design.

5 Results

5.1 Persistent participation responses

Figure 2 reports the event-study coefficients for the three turnout outcomes. The most consistent pattern is lower relative in-person turnout in court-linked areas at every subsequent election. Against the 2019 European reference, the estimates are −1.15 percentage points in the 2023 state rerun, −2.33 in the 2024 European election, −1.99 in the 2025 federal election, and −1.27 in the 2026 state election. All four are statistically significant under baseline precinct-clustered inference. The 2021 state and federal returns correspond to the same election day and underlying breakdown. The repeated negative contrasts across several electoral settings constitute the central behavioral finding: the relative in-person turnout deficit persists through 2026.

The magnitude is economically meaningful. The 2024 and 2025 estimates are around two percentage points, equivalent to approximately five percent of average in-person turnout in the harmonized panel. The contribution of postal voting is therefore important for understanding the associated changes in total participation.

Postal turnout shows a complementary pattern. Its coefficients are negative and imprecise in the concurrent 2021 returns, close to zero in the 2023 state rerun, and positive in the 2024 European and 2025 federal elections: 1.20 and 1.19 percentage points, respectively, both statistically significant. The 2026 estimate remains positive (0.43 points, SE 0.31), albeit less precisely estimated. Thus, court-linked areas exhibit relatively greater use of postal voting in two later elections alongside consistently lower in-person turnout. Most 2021 postal bal-

lots were cast before the election-day problems occurred; their contemporaneous coefficients instead provide context for differences already present at the time of the breakdown.

In 2024 and 2025, the positive postal coefficients offset approximately 51 and 60 percent of the corresponding in-person coefficients. These descriptive ratios summarize the opposing aggregate estimates. Since the two voting modes share the same electorate denominator and estimation sample, their coefficients add to the total-turnout estimates. These are −1.03 percentage points in 2023, −1.14 in 2024, −0.81 in 2025, and −0.84 in 2026: consistently negative and fairly stable in magnitude, though less precisely estimated than the in-person coefficients. The main pattern is therefore a persistent difference in voting-mode composition, accompanied by smaller negative estimates for total participation.

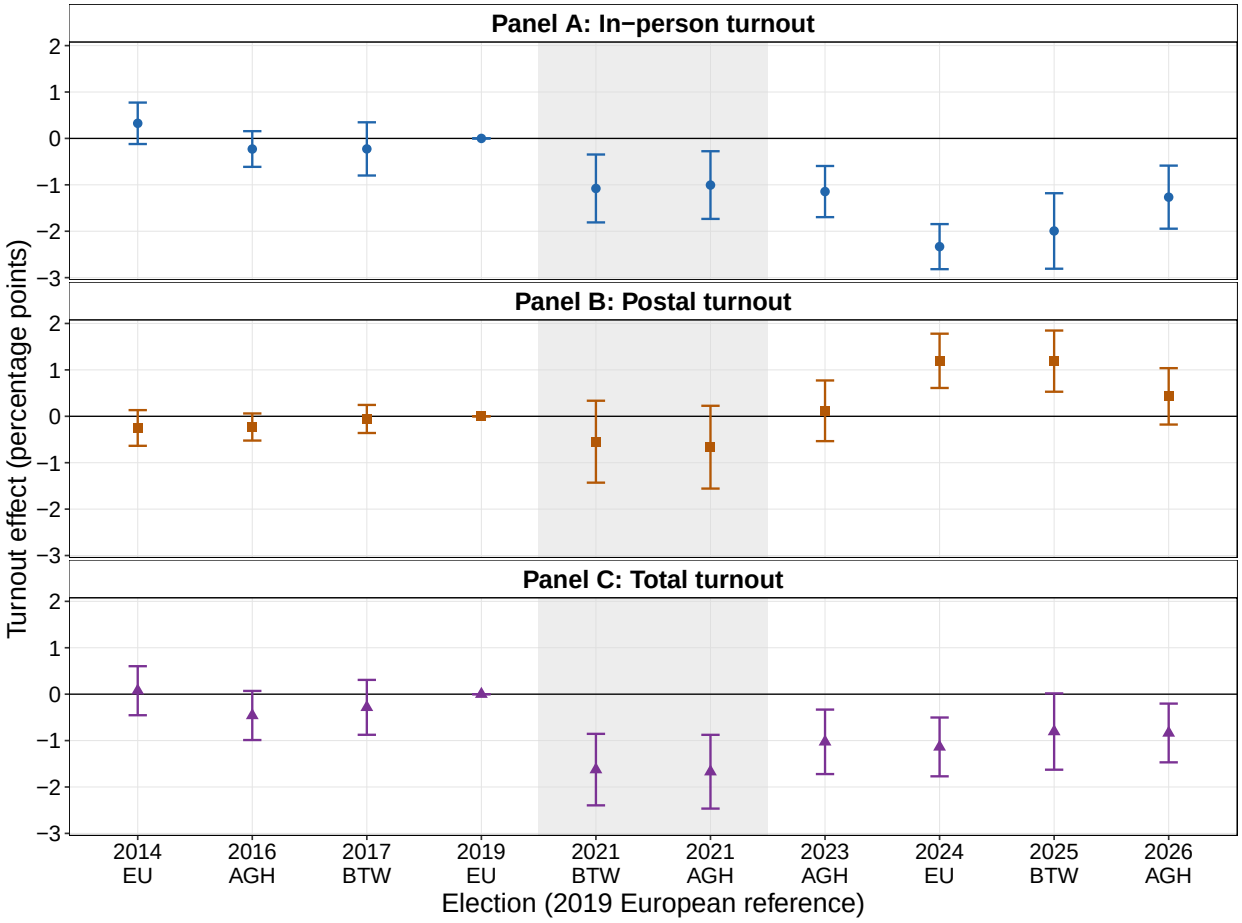
5.2 Voting-mode adjustment within election types

The common 2019 European-election reference provides a consistent benchmark for the event study. The 2014 and 2019 European elections make it possible to compare successive five-year intervals within European elections, while other within-family comparisons remain informative complements. The 2024 European election also expanded voting eligibility to ages 16–17, changing the electorate relative to 2019. The 2017–2025 federal comparison avoids this age-threshold change and compares two elections of the same type, both with an 18+ electorate. Table 1 also reports the 2023 and 2026 state-election comparisons relative to 2016 AGH; the complete within-family results appear in Appendix Table C2.

In the within-federal comparison, the in-person coefficient is −1.77 percentage points (SE 0.27), while the postal coefficient is +1.25 points (SE 0.37). The total-turnout estimate, their sum, is −0.52 points (SE 0.37), with an approximate 95% confidence interval from −1.25 to +0.21 points. The pronounced contrast between in-person and postal participation is the clearest within-election-family evidence of a persistent difference in local voting-mode composition. Postal turnout offsets much of the in-person difference in the point estimates, while total participation remains more uncertain. Both federal-election voting-mode contrasts remain statistically distinguishable from zero under district-level restricted Webb wild-cluster inference (in-person $p = 0.003$; postal $p = 0.008$), whereas the total-turnout contrast does not ($p = 0.202$; Appendix Table B5).

The 2023 state rerun shows an in-person deficit of −0.92 points (SE 0.26) relative to 2016 AGH, with less precise postal and total-turnout estimates. In 2026, the in-person estimate is again negative (−1.04 points, SE 0.29), while the total-turnout estimate is −0.38 (SE 0.31). The 2026 comparison spans the extension of the state franchise to 16- and 17-year-olds. Across both state and federal elections, the in-person differences are more clearly estimated

Figure 2: Turnout and voting-mode adjustment after the 2021 administrative failure



Notes: The figure presents event-study estimates based on Equation 4. Outcomes are in-person, postal, and total turnout, measured in percentage points of eligible voters. The reference election is the 2019 European election. The two 2021 outcomes are concurrent federal and state ballots from the same election day and reflect one underlying breakdown. The 2019 European election is normalized to zero; the 2014 European election provides a within-family pre-breakdown benchmark. Vertical bars show 95 percent confidence intervals based on standard errors clustered by 2021 postal precinct. State denotes Berlin AGH elections, Federal denotes Bundestag elections, and EU denotes European Parliament elections. The companion regression table is reported in Appendix Table B1.

Table 1: Voting-mode adjustment within comparable election types

Election	Reference	In-person turnout	Postal turnout	Total turnout
2023 AGH	2016 AGH	-0.917***	0.349	-0.568
		(0.259)	(0.345)	(0.353)
2025 Bundestag	2017 Bundestag	-1.768***	1.246***	-0.522
		(0.269)	(0.367)	(0.371)
2026 AGH	2016 AGH	-1.037***	0.661^{+}	-0.377
		(0.286)	(0.348)	(0.311)

Notes: Coefficients are percentage-point differences relative to the omitted election within each family: 2016 AGH for state elections and 2017 federal for federal elections. Standard errors (in parentheses) are clustered by 2021 postal precinct. Models include precinct and election-by-district fixed effects. The 2026 AGH electorate includes German citizens aged 16–17, unlike the 2016 reference. The complete results, including 2021 and European elections, appear in Appendix Table C2.

than the corresponding net changes in total participation.

5.3 Identification and robustness

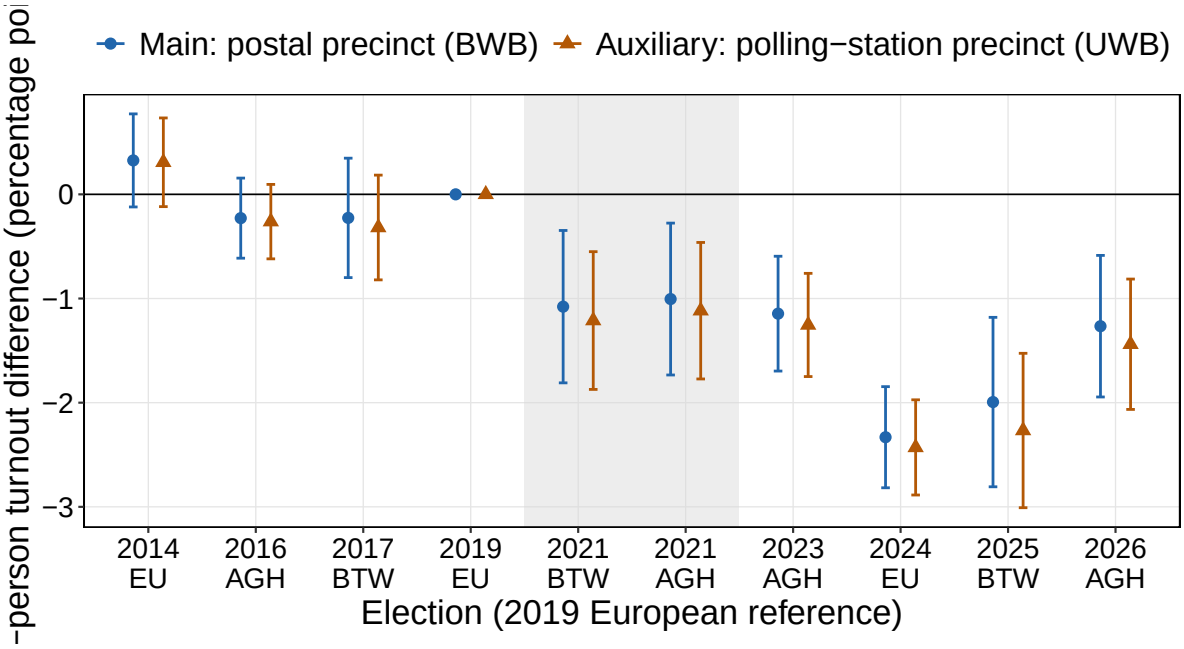
Baseline differences and counterfactual trends. The interpretation of these comparisons rests on conditional parallel trends within districts. Court-linked areas differed from clean comparison precincts before the breakdown: in 2019, treated precincts averaged about 1,884 residents versus 1,642 among controls, with differences in age and socioeconomic composition (Appendix Table B2). Precinct fixed effects absorb stable differences, and district-by-election effects account for shared local election shocks. The three pre-treatment elections offer useful evidence on differential trajectories: the 2014–2019 European-election change in relative in-person turnout is −0.33 points (SE 0.23), compared with −2.33 points (SE 0.25) between 2019 and 2024. The postal changes across these intervals are +0.25 (SE 0.20) and +1.20 (SE 0.30) points, respectively; total-turnout changes are −0.07 (SE 0.27) and −1.14 (SE 0.32). The 2016 total-turnout coefficient remains material relative to several later net-turnout estimates (Appendix Table B3; Appendix Table B4). The preferred interpretation is consequently an exposure-linked difference in local turnout trajectories under the stated within-district counterfactual-trends assumption.

The negative in-person coefficients persist with socioeconomic and baseline controls, exclusion of district 07, removal of adjacent comparison precincts, and matching procedures (Appendix Table C3; Appendix Table C4; Appendix Table C5; Appendix Table C6). Within-district placebo reassignment and district-clustered inference provide additional benchmarks for the strength and uncertainty of the estimated contrasts, as detailed in Section 4.2.

Geographic robustness. The harmonized postal-precinct panel combines returns from multiple reporting geographies. Socioeconomic and baseline controls preserve the negative post-2021 in-person pattern (Appendix Figure C1; Appendix Table C3). Concentration-index weighting and restriction to the full-panel k90 sample produce similar trajectories (Appendix Figure C5; Appendix Table C8). For the central 2017–2025 federal postal comparison, the estimate is +1.246 percentage points at baseline, +1.245 with crosswalk-HHI weighting, and +1.404 under the federal-pair k90 restriction, requiring at most two source units to cover 90% of destination mass in both federal elections (1,201 of 1,350 precincts; Appendix Table C1). The latter restriction changes the analysis population. These checks probe sensitivity to the existing allocation procedure; they do not supply an independent observation of postal votes at fixed residential geography.

The auxiliary analysis on original 2021 polling-station-precinct (UWB) geography provides an independent geographic comparison for in-person turnout. Figure 3 shows similar estimated declines on UWB and harmonized BWB geography, persisting through 2026 (Appendix Table C10). This agreement is especially informative for the in-person outcome, which is observed directly in the original polling-station returns. The harmonized panel also contains 18 precinct-election observations with total turnout above 100 percent, mostly in 2021. Excluding them leaves the terminal 2026 in-person coefficient unchanged at the reported precision (−1.266 percentage points, SE 0.346 in both samples; Appendix Table C11). The 2024 European and 2026 state elections expanded eligibility to ages 16–17, while the fixed allocation grid covers German citizens aged 18 and older. Election-specific official denominators incorporate the applicable franchise; the grid remains a common geographic allocation basis.

Figure 3: In-person turnout across geographic constructions



Notes: The figure compares the main event-study estimates on 2021 postal-precinct geography (BWB) with auxiliary estimates on 2021 polling-station-precinct geography (UWB). Both outcomes are in-person turnout in percentage points of eligible voters, with the 2019 European election as the reference. Symbols show point estimates and bars show 95% confidence intervals clustered at the respective 2021 unit. The shaded band marks the two concurrent 2021 ballots; they represent one disruption episode. State denotes Berlin AGH elections, Federal denotes Bundestag elections, and EU denotes European Parliament elections. The two constructions use different geographic units. Full auxiliary coefficients appear in Appendix Table C10.

5.4 Disruption families and the legal aftermath

The persistent turnout contrasts developed during a period that included the initial breakdown and its legal aftermath: the full 2023 state rerun and the geographically targeted 2024 federal rerun. The event-study coefficients summarize local participation trajectories across this institutional sequence.

The disruption-family comparisons add detail to this pattern. Direct-access and ballot-error-only precincts exhibit a significant difference in their 2024 in-person responses (-1.76 percentage points; $p = 0.004$), the clearest separation between these families across subsequent elections. Negative 2021 postal coefficients in the access-cost group attenuate after excluding district 07 (Appendix Figure D1; Appendix Table D1). Severity analyses offer a complementary exploratory comparison (Appendix Table D2). Taken together, these patterns are consistent with several possible behavioral responses to the breakdown and its aftermath.

6 Polling-Catchment Reorganization after 2021

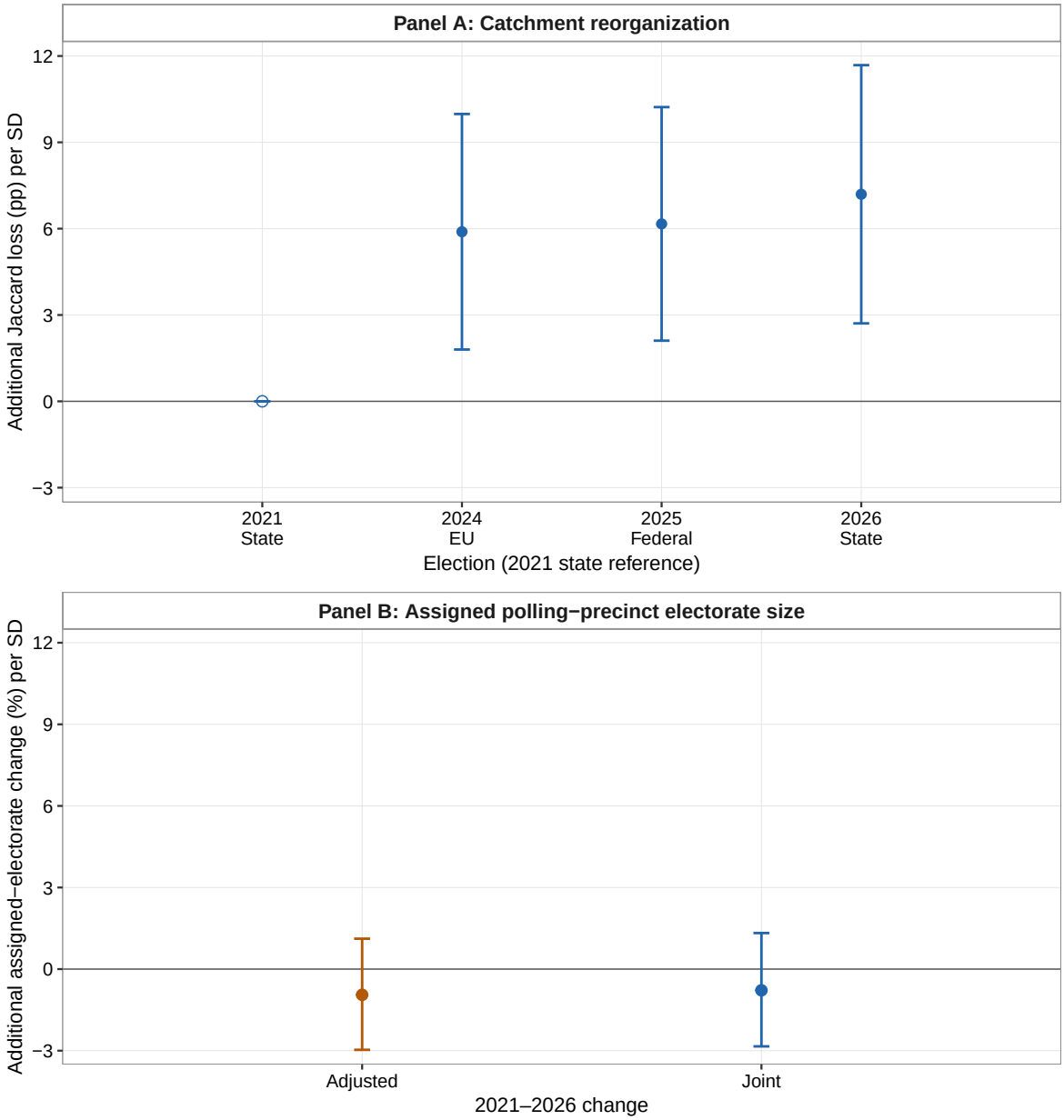
6.1 Catchment reorganization

The fixed-grid analysis measures how the in-person catchment serving each residential location changes relative to 2021. By 2026, substantial reorganization is observed in both exposure groups, with larger original precinct electorates generally associated with greater Jaccard loss. The raw levels reveal an important asymmetry. Among locations assigned to below-median-size 2021 precincts, average loss is 19.1% in court-linked areas, compared with 28.8% in clean controls. Among locations assigned to above-median-size precincts, the corresponding levels are almost identical: 43.3% and 43.6% (Appendix Table G1). Court-linked areas thus underwent considerably less reorganization at lower baseline electorate sizes, while the groups converged at higher sizes. This pattern makes the slope of reorganization with respect to initial electorate size the informative comparison.

The preferred joint specification estimates a 7.20-percentage-point steeper 2026 Jaccard-loss gradient per standard deviation of baseline original-precinct electorate size in court-linked areas, conditional on the other interacted baseline characteristics. The BWB21-clustered standard error is 2.29 percentage points; the district-clustered standard error is 2.14 points, and a restricted Webb wild-cluster test across Berlin's twelve districts yields $p = 0.017$ (Appendix Table G3). Descriptively, the steeper relationship coincides with less catchment change among initially smaller court-linked precincts and nearly identical mean losses among larger precincts. A nonlinear specification that interacts additional original-size terms with exposure estimates a +4.74-point coefficient on its linear workload-interaction component (Appendix Table G2). In that model the marginal slope difference varies with initial size, so the linear component represents a different parameter from the constant slope difference of the preferred specification. A binary measure of at least ten percent catchment loss also produces a positive interaction, while continuous Jaccard loss remains the preferred outcome. The population-weighted distribution is shown in Appendix Figure G1.

Figure 4 displays the catchment-reorganization estimates alongside a separate outcome based on the electorate assigned to in-person precincts. These measures describe two complementary aspects of election administration: the composition of residential catchments and the number of eligible voters they serve. Alternative initial-size measures, threshold outcomes, and overlap normalizations provide further evidence on the sensitivity of the reorganization gradient to its geographic definition (Appendix Figure G2; Appendix Figure G3; Appendix Table G2).

Figure 4: Polling-catchment reorganization and assigned precinct electorates



Notes: Panel A shows interaction estimates for continuous UWB catchment Jaccard loss from the joint fixed-grid specification, relative to the 2021 administrative reference. Panel B shows 2021–2026 interactions for assigned electorate size in the UWBs serving fixed residential locations, aggregated to BWB21 areas in adjusted and joint models. Values are converted to percentage points or approximate percent workload changes for display. Capped bars show 95% confidence intervals using BWB21-clustered standard errors. Model specifications and district-level inference are detailed in Appendix Section G.

6.2 Assigned precinct electorate size

The second administrative outcome follows the electorate assigned to the election-year UWB serving each fixed residential location in 2021 and 2026. Each cell is linked to its actual UWB and that precinct's official electorate in the respective election; the values are then aggregated to the original BWB21 areas using fixed residential-population weights. The resulting change in assigned electorate size serves as a proxy for administrative workload. In the adjusted and joint specifications, the exposure-by-baseline-electorate interactions are small and negative, at approximately −0.95 and −0.78 percent per standard deviation, respectively. They are estimated less precisely than the catchment-reorganization gradient, including under district-clustered and Webb wild-cluster inference (Appendix Table G4). The analysis retains all 1,350 BWB21 areas after auditing three legitimate zero-electorate residential cells. The underlying administrative data also show descriptive convergence in assigned electorate size (Appendix Figure G4). The 2026 state electorate includes newly eligible 16–17-year-olds, so the change combines geographic reassignment with evolution of the eligible population.

Table 2: Polling-catchment reorganization and assigned electorate size

Outcome and specification	Interaction	SE: BWB21	SE: district	*N*
Panel A. Catchment reorganization (Jaccard percentage points)				
Continuous Jaccard loss, joint	+7.20	(2.29)	(2.14)	36,594
Panel B. Change in assigned polling-precinct electorate size (approx. percent)				
Adjusted	-0.95	(1.05)	(1.59)	1,350
Joint	-0.78	(1.07)	(1.48)	1,350

Notes: The table reports the 2026 disruption-by-standardized-baseline-electorate interactions. The two panels measure distinct outcomes: Jaccard catchment loss in percentage points and log changes in assigned precinct electorate size in approximate percentage units. Standard errors are conventional cluster-robust estimates at BWB21 and Berlin district levels. The original estimation uses fixed residential-population weights and models specified in Section 4.3. Full models, binary reorganization, and Webb sensitivity are in Appendix Section G. The interactions describe exposure-specific gradients conditional on the included baseline measures.

6.3 Voter and institutional adjustment

The participation and administrative results reveal a useful sequence. The 2023 state rerun retained the 2021 polling-catchment geography, yet court-linked areas already exhibited a negative relative in-person turnout contrast. Subsequent elections featured persistent voting-mode differences alongside changes in the geography of in-person voting. By 2026, catchment reorganization rose more steeply with original precinct electorate size in court-linked areas,

particularly because initially smaller exposed precincts experienced less reorganization; the assigned-electorate results describe a separate dimension of this administrative evolution. Together, the analyses show how the use of voting channels and the territorial organization of elections can follow distinct trajectories after the same institutional failure.

Evidence on original polling-location sites is reported in Appendix Section E, while survey responses document the public salience of the breakdown (Appendix Section F).

7 Conclusion

Berlin's 2021 electoral breakdown and its legal aftermath provide an opportunity to examine persistent differences in how local electorates participate. Court-linked areas exhibit more negative changes in in-person turnout through 2026, alongside relatively higher postal turnout in the 2024 European and 2025 federal elections. Between the 2017 and 2025 federal elections, a -1.77-point in-person contrast is accompanied by a $+1.25$-point postal contrast, leaving a smaller and less precisely estimated -0.52-point change in total turnout. The principal finding is a persistent difference in voting-mode composition across subsequent elections.

The fixed-grid analysis adds a complementary institutional perspective. By 2026, catchment reorganization is more strongly related to initial precinct electorate size in court-linked areas, with less reorganization among initially smaller exposed precincts and nearly identical mean levels among larger precincts. The preferred conditional gradient is $+7.20$ percentage points of Jaccard loss per standard deviation of baseline electorate size. Alternative overlap measures and nonlinear-size specifications illuminate the sensitivity of that relationship to the geographic and functional-form choices. The separately measured changes in assigned electorate size are smaller and estimated less precisely.

The two analyses speak to a broader political-economy distinction. A negative relative in-person turnout contrast was already visible in the 2023 state rerun, before the later catchment reorganization documented here. Following the same institutional episode across elections thus reveals both the evolution of local voting behavior and subsequent changes in the geography of electoral service provision. Assessing electoral reliability requires distinguishing the opportunity to vote, the channel citizens use, and the organization through which that opportunity is delivered.

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A Electoral institutions and spatial measurement

Exhibit conventions. EU denotes European Parliament elections, AGH the Berlin state elections, and BTW Bundestag elections. Unless stated otherwise, regression entries are coefficients with clustered standard errors in parentheses; $^{+}$ denotes $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$. Wild-cluster bootstrap p-values are reported separately where used.

German federal and Berlin state elections use personalized proportional representation. Citizens receive election notifications identifying their assigned polling location and may request postal ballots for each election. Voting by post is an established option, but requesting, completing, and returning the documents must take place within election-specific deadlines. These administrative distinctions motivate separate treatment of election-day and postal ballot counts in the empirical panel.

The treatment taxonomy draws on two partially overlapping court sources. Federal-linked exposure defines the principal treatment contrast; state-linked records additionally classify documented queues, interruptions, missing ballots, and ballot errors. A unit appearing only in the state record is not included as a clean federal control. The descriptive counts below establish how much the primary treatment and the state-record mechanism groups overlap. They do not establish that every voter residing in a classified precinct experienced the reported event.

The classification maps original 2021 polling-station precincts (UWBs) to postal precincts (BWB21) using their official 2021 assignments, rather than the population-weighted crosswalk for electoral returns. For each BWB21 area, federal exposure is the unweighted proportion of constituent UWBs appearing in the federal record. The state record provides both an indicator for any constituent UWB with a documented incident and an electorate-weighted exposed-UWB share. A BWB21 area is court-linked treated if its federal share exceeds 75% without a state-record incident, or exceeds 20% with at least one state-record incident. Clean controls have federal and state exposure shares below 1%; remaining areas, including state-only and intermediate-share cases, are excluded from the preferred comparison. This fixed classification is reproduced in the dataset-building script.

The panel combines results originally reported on separate polling-station and postal-precinct geographies. For each source precinct s and 2021 target precinct i, a population-weighted crosswalk assigns a nonnegative share w_{si}, with $\sum_i w_{si} = 1$ for each source. A source-level count V_{st} is allocated as $\widetilde{V}_{it} = \sum_s w_{si} V_{st}$, using distinct crosswalks for in-person and postal sources. Turnout rates use correspondingly allocated official election-specific registered-voter counts. By construction, source counts are preserved when the weights sum

to one. Conservation does not independently validate the distribution of allocated counts within Berlin.

For the 2014 European election, the corrected official returns report 1,709 in-person precincts and 529 distinct postal reporting districts. The historical polling-precinct polygon layer identifies each in-person precinct by `STI` and provides its associated postal district in `BRFW`. Dissolving the 1,709 polygons by `BRFW` reconstructs all 529 postal reporting areas, whose identifiers match the separate postal-return records; no original source units are unmatched. In-person and postal counts are then allocated independently from their respective 2014 source areas to the fixed BWB21 geography. This validates the source-area linkage, not the precision of the subsequent population-weighted interpolation. A direct source-to-target audit recovers the complete 2014 in-person and postal counts: 841,916 in-person voters, 335,916 postal voters, and 2,519,758 eligible voters. The 2014 allocation-quality audit covers all 294 court-linked and 1,056 clean-control BWB21 units; its mass-fraction and source-unit diagnostics nevertheless indicate differences in interpolation concentration across exposure groups. The 2014 postal crosswalk is more diffuse than its 2019 counterpart (median effective destination counts of 3.46 and 2.74), which is relevant when interpreting the 2014–2019 pre-period diagnostic.

The underlying 2022 grid measures German citizens aged 18 and older. This is a constant proxy for the geographical distribution of eligible voters, not a claim that all election-specific franchises are identical. The 2024 European and 2026 state franchises include eligible 16–17-year-olds, while the former also includes non-German EU citizens. Interpolation precision may vary with the extent of boundary overlap and spatial population composition. Appendix crosswalk tests therefore report both mapping complexity and regression sensitivity. A true 16+-matched sensitivity would require verified fine-resolution population counts; none are imputed. In figure axes and regression tables, AGH denotes the Berlin state election, BTW the federal Bundestag election, and EU the European Parliament election. The displays first document source exposure, summary statistics, and spatial allocation, followed by the detailed state-record disruption taxonomy.

Table A1: Source-overlap counts

Source category	Postal precincts	Share
No court record	1056	0.701
Federal record only	17	0.011
State record only	157	0.104
Federal and state records	277	0.184

Notes: The table reports counts of 2021 postal precincts by federal and state court-record source category. Clean controls are precincts not mentioned in either record. The corresponding figure is Figure A1.

Table A2: Harmonized returns and preferred estimation sample

Analysis population	2021 postal precincts	Elections	Area–election observations
All harmonized election returns	1,507	10	15,070
Court-linked treated areas	294	10	2,940
Clean comparison areas	1,056	10	10,560
Excluded from preferred comparison	157	10	1,570
Preferred estimation sample	1,350	10	13,500

Notes: Areas are defined on the fixed 2021 postal-precinct geography. Each class contains ten elections from 2014 through 2026. Court-linked exposure follows the federal record and the stated threshold classification. The 157 state-only record areas are excluded from the preferred federal-record comparison, not relabeled as clean controls; they are the excluded category in this accounting rather than a separate fifth map class. Main regressions retain 294 treated and 1,056 clean comparison areas and have 13,500 observations per turnout outcome.

Table A3: Summary statistics: source and preferred analysis populations

Variable	Source N	Source mean	Source SD	Preferred N	Preferred mean	Preferred SD
AfD votes (eligible-voter share)	15070	7.346	4.696	13500	7.354	4.653
AfD vote share (valid votes)	15070	10.850	6.842	13500	10.834	6.751
CDU votes (eligible-voter share)	15070	13.418	7.444	13500	13.702	7.556
CDU vote share (valid votes)	15070	19.262	9.486	13500	19.615	9.573
Eligible voters	15070	1644.988	609.819	13500	1639.790	609.183
FDP votes (eligible-voter share)	15070	3.975	3.062	13500	4.027	3.098
FDP vote share (valid votes)	15070	5.459	3.656	13500	5.517	3.685
Green votes (eligible-voter share)	15070	12.996	8.128	13500	12.954	8.063
Green vote share (valid votes)	15070	18.364	10.476	13500	18.292	10.394
Left votes (eligible-voter share)	15070	10.409	7.212	13500	10.200	7.146
Left vote share (valid votes)	15070	15.061	9.796	13500	14.750	9.746
In-person turnout (eligible-voter share)	15070	41.887	7.384	13500	41.894	7.395
Postal turnout (eligible-voter share)	15070	27.267	9.689	13500	27.379	9.725
SPD votes (eligible-voter share)	15070	12.344	4.170	13500	12.443	4.186
SPD vote share (valid votes)	15070	18.011	5.505	13500	18.119	5.501
Total turnout (eligible-voter share)	15070	69.154	13.035	13500	69.273	13.048
Residents	15070	1696.469	350.534	13500	1694.892	350.439
Age 18-25 share	15070	7.826	2.446	13500	7.834	2.453
Age 25-35 share	15070	16.158	7.047	13500	15.860	6.983
Age 35-45 share	15070	16.527	5.741	13500	16.253	5.590
Age 45-60 share	15070	24.903	4.681	13500	24.992	4.701
Age 60-70 share	15070	14.264	4.020	13500	14.403	3.944
Age 65+ share	15070	20.451	8.929	13500	20.826	8.923
Age 70+ share	15070	20.321	9.608	13500	20.658	9.610
Migration background (share)	15070	13.844	8.104	13500	13.758	8.066
SGB II recipients (share)	15070	13.272	9.413	13500	13.146	9.438
Foreign citizens (share)	15070	19.224	10.131	13500	18.910	10.099

Notes: The table reports summary statistics for the analysis panel. Variables include turnout outcomes, selected party vote shares, and socio-demographic controls. The *source* columns describe the ten-election (2014–2026) working panel, including precincts excluded from the preferred comparison. The *preferred* columns describe court-documented federal exposure versus precincts without either court record. Variable-specific N differs when data are missing. The preferred balanced turnout regressions include 13,500 precinct-election observations.

Table A4: Crosswalk quality summary

Election	Geometry	Source units	Mean weight sum	Min weight sum	Max targets/source	Mean targets/source
2014 European	Polling-station precinct to BWB21	1709	1	1	17	5.365
2014 European	Postal precinct to BWB21	529	1	1	24	10.338
2016 State	Polling-station precinct to BWB21	1779	1	1	18	4.997
2016 State	Postal precinct to BWB21	653	1	1	28	8.763
2017 Federal	Polling-station precinct to BWB21	1779	1	1	16	4.187
2017 Federal	Postal precinct to BWB21	660	1	1	22	7.771
2019 European	Polling-station precinct to BWB21	1800	1	1	16	4.173
2019 European	Postal precinct to BWB21	718	1	1	19	7.400
2024 European	Polling-station precinct to BWB21	2220	1	1	7	1.613
2024 European	Postal precinct to BWB21	1205	1	1	8	2.390
2025 Federal	Polling-station precinct to BWB21	2323	1	1	6	1.467
2025 Federal	Postal precinct to BWB21	1275	1	1	6	2.107
2026 State	Polling-station precinct to BWB21	2542	1	1	15	4.366
2026 State	Postal precinct to BWB21	1572	1	1	17	5.574

Notes: The table summarizes population-weighted crosswalks from historical election geographies to 2021 postal precincts. It documents the harmonization process used to place in-person, postal, and total turnout on a common geography.

Table A5: 2014 European election: source counts and allocation conservation

Original reporting unit	Source areas	Eligible voters	Voters	Valid votes
In-person	1,709	2,519,758	841,916	827,763
Postal	529	0	335,916	331,903
Source-to-target allocation	All source IDs matched; all original totals conserved			

Notes: Corrected official results contain 1,709 in-person reporting units and 529 distinct postal reporting units. The historical `STI` and `BRFW` identifiers link each source unit to its original reporting geography. Postal reporting rows have no separate eligible-electorate count; the official electorate is recorded in the in-person rows. All four source quantities (voters, eligible voters, valid votes and invalid votes) are preserved when mapped to the 2021 geography. Exact conservation establishes accounting consistency, not the absence of spatial interpolation error.

Table A6: Allocation concentration across exposure groups, 2014 and 2019

Year / geography	Exposure	Areas	Median fraction	Weighted fraction	Sources/target
2014 postal	Clean control	1056	0.246	0.286	3.54
2014 postal	Court-linked	294	0.294	0.324	3.93
2019 postal	Clean control	1056	0.321	0.373	3.52
2019 postal	Court-linked	294	0.383	0.448	3.58
2014 in-person	Clean control	1056	0.594	0.656	5.89
2014 in-person	Court-linked	294	0.770	0.761	6.74
2019 in-person	Clean control	1056	0.663	0.703	4.95
2019 in-person	Court-linked	294	0.822	0.802	5.03

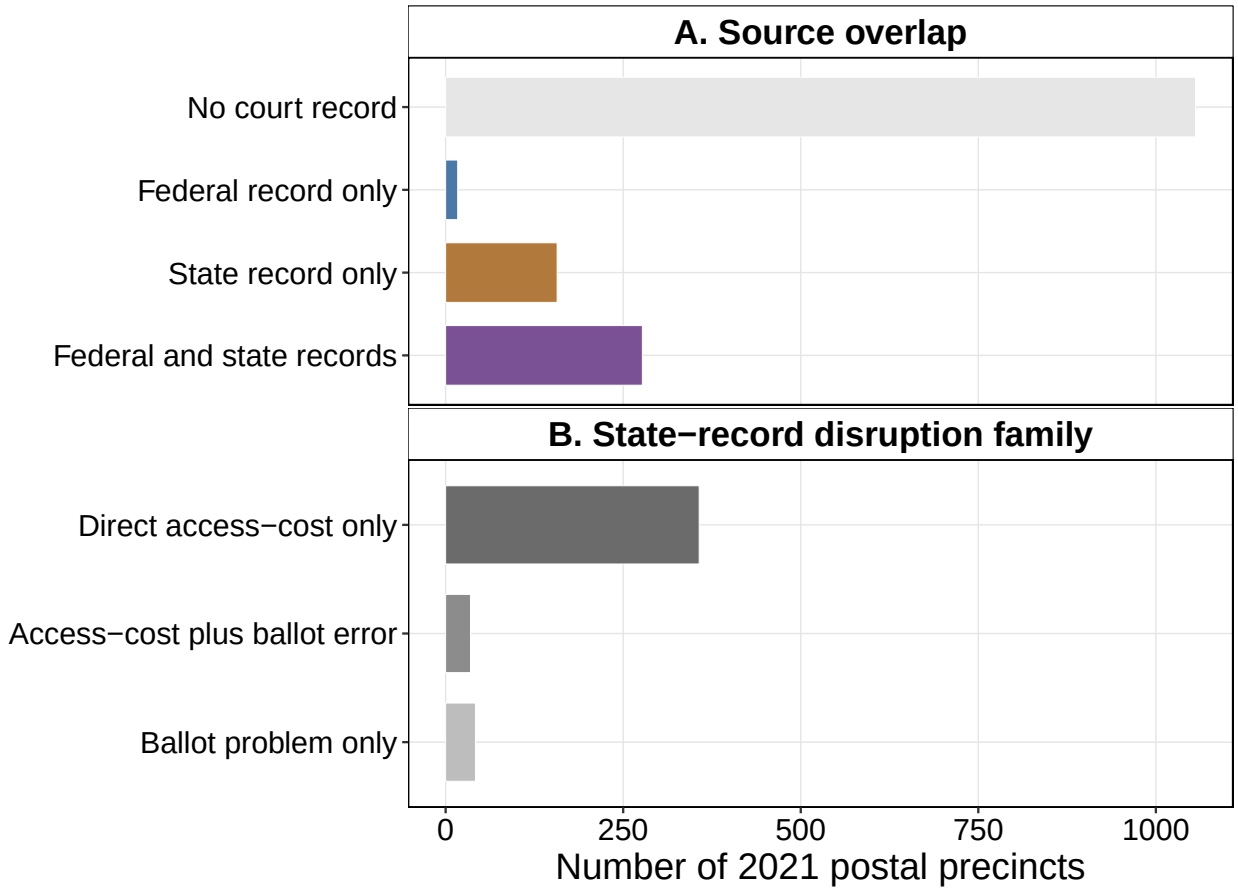
Notes: Postal and in-person refer to the historical reporting units used to allocate votes and electorates to fixed 2021 BWB21 target areas. For each target, the source fraction is the grid-population-weighted mean of the proportion of each contributing source unit allocated to that target. The table reports the median across targets and the corresponding target-population-weighted mean. Larger fractions generally indicate more concentrated source-to-target allocation, while the number of contributing source units gives a separate fragmentation measure. Neither statistic is a direct measure of actual allocation error. Groups include 294 federally court-linked and 1,056 clean-control areas and omit the 157 state-only excluded areas.

Table A7: Audit of harmonized turnout above 100 percent

outcome	Observations	Observations above 100 percent	Share above 100 percent	Maximum
In-person turnout (eligible-voter share)	15070	0	0.000	68.818
Postal turnout (eligible-voter share)	15070	0	0.000	86.934
Total turnout (eligible-voter share)	15070	18	0.001	126.071

Notes: The table counts harmonized precinct-election observations with turnout above 100 percent. These observations can arise when election-specific votes and electorate denominators are allocated from different source geometries. The positive-turnout sample screen is applied after harmonization and imposes no 100% upper cutoff; the preferred regressions retain eligible observations with rates above 100%, while a separate exclusion sensitivity examines their influence. The geographic checks are reported in Figure C5 and table C8.

Figure A1: Source overlap and state-record disruption families



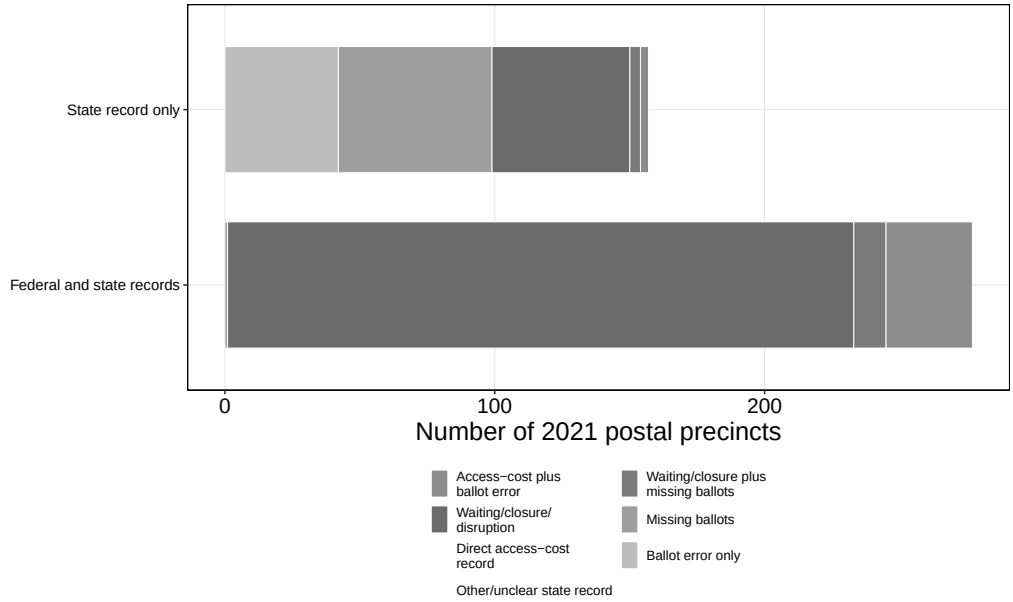
Notes: Panel A counts 2021 postal precincts by overlap between federal and state court-linked records. Panel B counts state-record precincts by broad disruption family. Direct access/cost disruptions include waiting, voting after closing, closures or interruptions, and missing ballots. Ballot-problem-only records capture ballot errors without direct access/cost disruption. The companion tables are Tables A1 and A8.

Table A8: State-record disruption-family counts

State-record disruption family	Postal precincts	Share
Direct access-cost only	357	0.823
Access-cost plus ballot error	35	0.081
Ballot problem only	42	0.097

Notes: The table reports counts of state-record precincts by broad disruption family. Missing ballots are coded as access/cost disruptions because they prevent or delay normal in-person voting.

Figure A2: Full state-record disruption taxonomy



Notes: The figure reports the detailed state-record disruption taxonomy. It is descriptive and clarifies the construction of the disruption-family comparisons in Appendix Figure D1. The corresponding counts are reported in Table A9.

Table A9: Full state-record disruption taxonomy counts

Source category	Detailed state-record category	Broad family	Postal precincts	Share within source
Federal and state records	Missing ballots	Direct access-cost only	1	0.004
Federal and state records	Waiting/closure plus missing ballots	Direct access-cost only	12	0.043
Federal and state records	Waiting/closure/disruption	Direct access-cost only	232	0.838
Federal and state records	Access-cost plus ballot error	Access-cost plus ballot error	32	0.116
State record only	Missing ballots	Direct access-cost only	57	0.363
State record only	Waiting/closure plus missing ballots	Direct access-cost only	4	0.025
State record only	Waiting/closure/disruption	Direct access-cost only	51	0.325
State record only	Access-cost plus ballot error	Access-cost plus ballot error	3	0.019
State record only	Ballot error only	Ballot problem only	42	0.268

Notes: The table reports the detailed state-record disruption taxonomy underlying Figure A2.

B Identification and baseline diagnostics

The preferred design compares changes in court-linked treated and clean-control postal precincts relative to the 2019 European election, conditional on precinct and election-by-district fixed effects. Exposure is documented rather than randomized. The descriptive turnout gaps, baseline estimates, and 2019 balance statistics document the initial empirical comparison. Pre-treatment coefficients, within-district treatment-reassignment diagnostics, and alternative inference assess the sensitivity of the comparison.

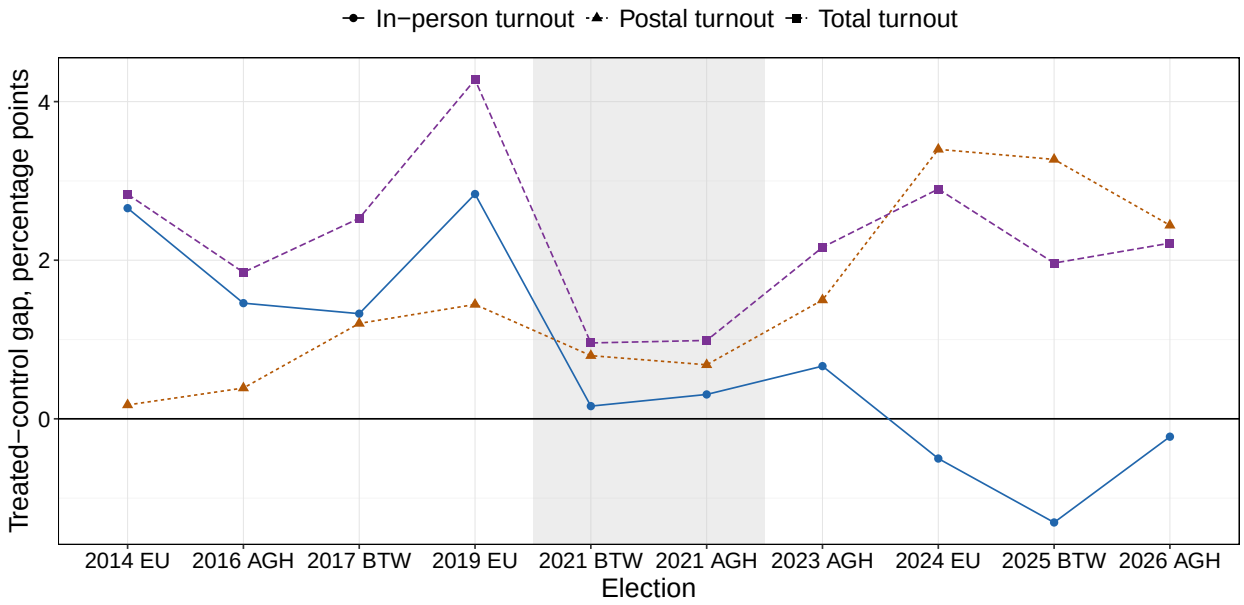
Three non-reference pre-treatment elections are observed, including the 2014 European election, which permits a five-year within-family comparison with the 2019 reference. The pre-period evidence improves comparability but remains limited to a single pre-breakdown interval within European elections. The 2014 crosswalk and postal-source construction should be assessed alongside the geographical quality table in Appendix A. The historical polygon layer links each in-person unit (`STI`) to a distinct postal reporting district (`BRFW`); these are separate identifiers, and the 529 postal areas are reconstructed by dissolving the in-person polygons according to their postal-district assignments. Placebo treatment labels are reassigned within districts while preserving district-specific treated counts, providing a reference distribution rather than design-based inference from an actual randomized assignment. The table comparing post-treatment coefficients with the largest observed pre-treatment coefficient is a descriptive calibration and not a formal bound on confounding or an HonestDiD identified set.

Table B1: Main specification: turnout levels

Election	(1) In-person turnout	(2) Postal turnout	(3) Total turnout
2014 EU	0.326	-0.251	0.074
	(0.228)	(0.196)	(0.269)
2016 AGH	-0.228	-0.231	-0.459^{+}
	(0.196)	(0.149)	(0.270)
2017 federal	-0.226	-0.058	-0.284
	(0.292)	(0.154)	(0.301)
2021 federal	-1.078**	-0.547	-1.625***
	(0.373)	(0.450)	(0.393)
2021 AGH	-1.005**	-0.666	-1.671***
	(0.372)	(0.455)	(0.405)
2023 AGH	-1.145***	0.118	-1.027**
	(0.281)	(0.333)	(0.354)
2024 European	-2.332***	1.195***	-1.137***
	(0.247)	(0.298)	(0.323)
2025 federal	-1.994***	1.188***	-0.806^{+}
	(0.415)	(0.336)	(0.419)
2026 AGH	-1.266***	0.430	-0.836**
	(0.346)	(0.310)	(0.323)
Fit statistics			
Observations	13,500	13,500	13,500
Fixed effects			
Precinct	Yes	Yes	Yes
Election × District	Yes	Yes	Yes

Notes: The table presents event-study results based on Equation 4. Columns (1), (2), and (3) report in-person, postal, and total turnout, respectively. Outcomes are measured in percentage points of eligible voters. The reference election is the 2019 European election. All specifications include 2021 postal-precinct fixed effects and election-by-district fixed effects. The corresponding figure is Figure 2.

Figure B1: Treated-control turnout gaps by election and voting mode



Notes: The figure plots treated-control differences in in-person, postal, and total turnout across elections. Turnout is measured as a percentage of eligible voters. The treatment is the main federal court-linked exposure indicator. The figure is descriptive and does not condition on postal-precinct fixed effects or election-by-district fixed effects.

Table B2: Pre-treatment characteristics by court-linked exposure, 2019 EU

Characteristic	Treated mean	Control mean	Std. difference	Treated N	Control N
Residents	1883.577	1642.360	0.743	294	1056
Foreign-citizen share	20.283	18.528	0.178	294	1056
Migration-background share	13.622	13.796	-0.022	294	1056
SGB II share	11.413	13.628	-0.242	294	1056
Age 18-25 share	7.806	7.842	-0.014	294	1056
Age 25-35 share	16.744	15.613	0.170	294	1056
Age 35-45 share	17.817	15.817	0.354	294	1056
Age 45-60 share	26.475	24.579	0.403	294	1056
Age 60-70 share	13.127	14.758	-0.422	294	1056
Age 65+ share	18.075	21.591	-0.399	294	1056
Age 70+ share	18.032	21.390	-0.351	294	1056

Notes: The table reports 2019 European-election pre-treatment characteristics for the preferred treated and clean-control sample. Standardized differences equal the treated-minus-control mean divided by the square root of the average group variance. The comparisons are descriptive; district-adjusted treatment-prediction regressions are retained separately in the replication outputs.

Table B3: Pre-treatment event-study coefficients

Outcome	Election	Estimate	Std. error	95% CI lower	95% CI upper	p-value
In-person turnout	2014 EU	0.326	0.228	-0.121	0.772	0.153
In-person turnout	2016 AGH	-0.228	0.196	-0.612	0.156	0.244
In-person turnout	2017 federal	-0.226	0.292	-0.799	0.347	0.439
Postal turnout	2014 EU	-0.251	0.196	-0.636	0.133	0.200
Postal turnout	2016 AGH	-0.231	0.149	-0.523	0.061	0.121
Postal turnout	2017 federal	-0.058	0.154	-0.360	0.245	0.709
Total turnout	2014 EU	0.074	0.269	-0.453	0.602	0.783
Total turnout	2016 AGH	-0.459	0.270	-0.988	0.069	0.089
Total turnout	2017 federal	-0.284	0.301	-0.874	0.307	0.347

Notes: The table reports pre-treatment coefficients for the main event-study design. The reference election is the 2019 European election.

Table B4: European-election pre- and post-breakdown five-year changes

Turnout	2014–19 change	2019–24 change	Difference between changes
In-person	-0.326 (0.228)	-2.332 (0.247)	-2.006 (0.349)
Postal	+0.251 (0.196)	+1.195 (0.298)	+0.943 (0.365)
Total	-0.074 (0.269)	-1.137 (0.323)	-1.062 (0.447)

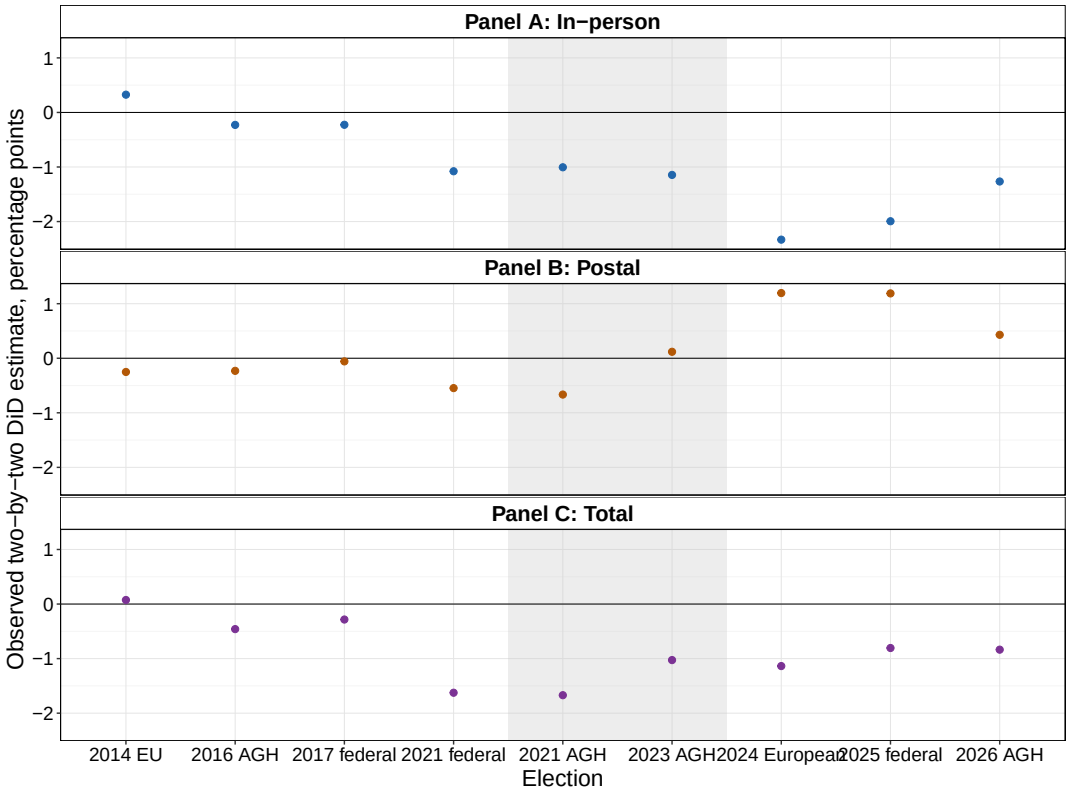
Notes: All entries are changes in the court-linked minus control turnout difference (percentage points). The 2014–2019 change is the negative of the EU14 coefficient in the 2019-normalized event study; the 2019–2024 change is the EU24 coefficient. The final column compares these two five-year changes and uses their full joint precinct-clustered covariance matrix. The 2024 franchise covers ages 16–17, unlike the 2014 and 2019 elections. The 2014 source-geography audit is also reported in the replication outputs.

Table B5: Few-cluster inference for the 2017–2025 federal turnout comparison

Turnout	Estimate	Precinct SE	District SE	Webb p	N
In-person	-1.768	0.269	0.400	0.003	2,700
Postal	+1.246	0.367	0.450	0.008	2,700
Total	-0.522	0.371	0.369	0.202	2,700

Notes: Point estimates compare 2017 and 2025 federal elections using precinct and district-by-election fixed effects. District-clustered SEs use Berlin's twelve districts. Webb p-values use 1,999 restricted wild-cluster draws and a two-sided null of no treated–control change, after verification that the equivalent district-adjusted first-difference regression reproduces the FE coefficient. The original precinct-clustered SEs are shown alongside them.

Figure B2: Within-district treatment-reassignment diagnostics



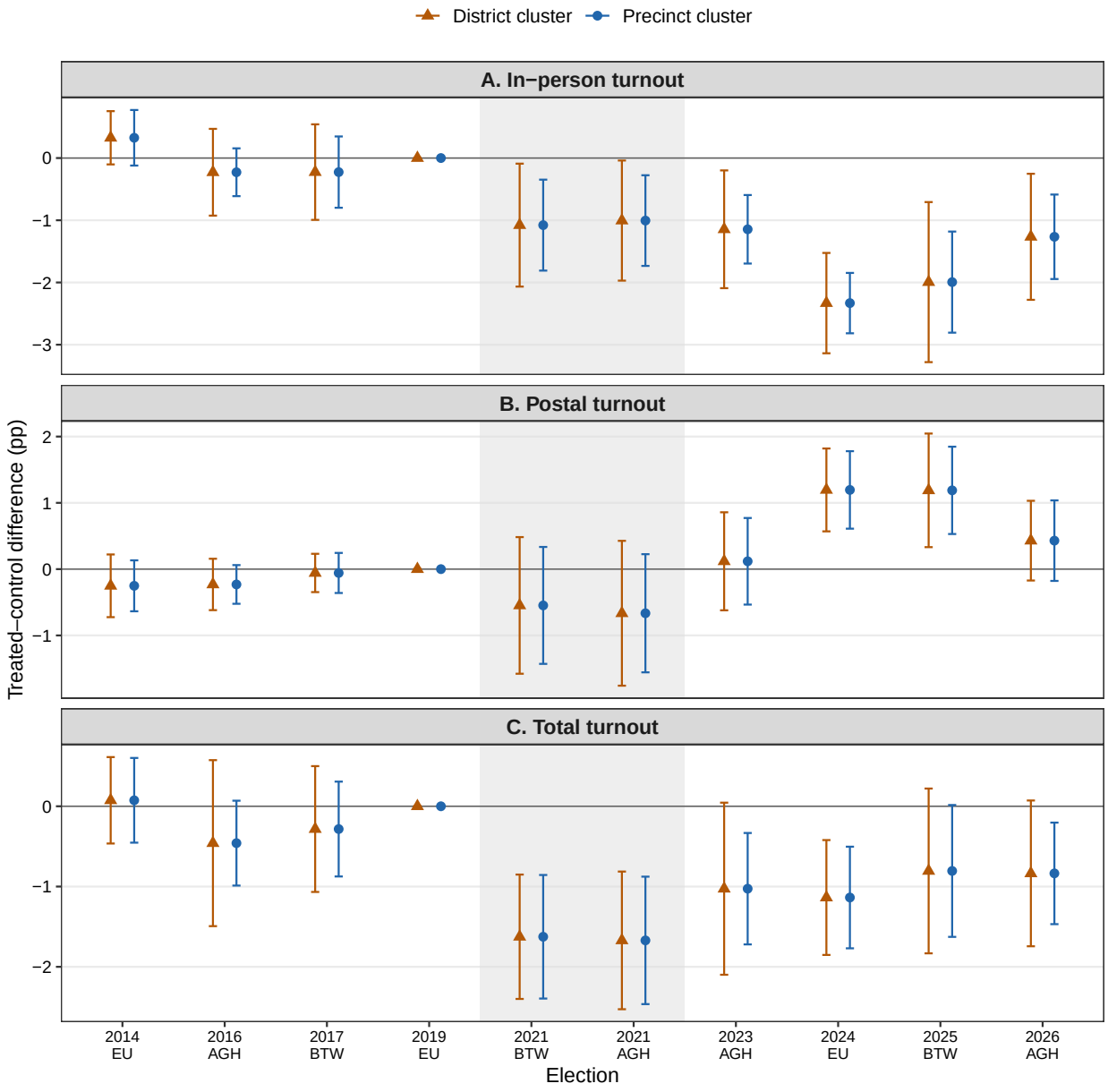
Notes: The figure compares observed event-specific estimates with a reference distribution generated by reassigning treatment labels while holding the number treated within each district fixed. It does not treat actual court-linked exposure as randomly assigned. Comparisons are made by election and voting mode relative to 2019 EU; see Table B6.

Table B6: Within-district treatment-reassignment diagnostics by election and voting mode

Election vs. 2019 reference	In-person turnout	Postal turnout	Total turnout
2014 EU	0.326 [p=0.135]	-0.251 [p=0.183]	0.074 [p=0.785]
2016 AGH	-0.228 [p=0.279]	-0.231 [p=0.094]	-0.459 [p=0.091]
2017 federal	-0.226 [p=0.440]	-0.058 [p=0.710]	-0.284 [p=0.332]
2021 federal	-1.078 [p=0.003]	-0.547 [p=0.190]	-1.625 [p=0.001]
2021 AGH	-1.005 [p=0.007]	-0.666 [p=0.120]	-1.671 [p=0.001]
2023 AGH	-1.145 [p=0.001]	0.118 [p=0.732]	-1.027 [p=0.006]
2024 European	-2.332 [p=0.001]	1.195 [p=0.001]	-1.137 [p=0.001]
2025 federal	-1.994 [p=0.001]	1.188 [p=0.001]	-0.806 [p=0.070]
2026 AGH	-1.266 [p=0.001]	0.430 [p=0.170]	-0.836 [p=0.012]

Notes: Each row compares the listed election with 2019 EU in a two-period difference-in-differences specification. Placebo treatment labels are reassigned among 2021 postal precincts, preserving the number treated within each district. Reported reference-distribution probabilities describe the position of the observed estimate under this reassignment exercise, rather than a randomized-assignment test. See Figure B2.

Figure B3: Precinct- and district-clustered uncertainty



Notes: The figure compares 2021 postal-precinct clustering with clustering by Berlin district (12 clusters). The reference election is the 2019 European election. The companion table is Table B8.

Table B7: Descriptive calibration using pre-treatment coefficients

Outcome	Election	Estimate	Std. error	Max. absolute pre-period coefficient	Post/pre magnitude ratio
In-person turnout	2021 federal	-1.078	0.373	0.326	3.310
In-person turnout	2021 AGH	-1.005	0.372	0.326	3.085
In-person turnout	2023 AGH	-1.145	0.281	0.326	3.516
In-person turnout	2024 European	-2.332	0.247	0.326	7.157
In-person turnout	2025 federal	-1.994	0.415	0.326	6.121
In-person turnout	2026 AGH	-1.266	0.346	0.326	3.885
Postal turnout	2021 federal	-0.547	0.450	0.251	2.176
Postal turnout	2021 AGH	-0.666	0.455	0.251	2.647
Postal turnout	2023 AGH	0.118	0.333	0.251	0.471
Postal turnout	2024 European	1.195	0.298	0.251	4.752
Postal turnout	2025 federal	1.188	0.336	0.251	4.726
Postal turnout	2026 AGH	0.430	0.310	0.251	1.709
Total turnout	2021 federal	-1.625	0.393	0.459	3.539
Total turnout	2021 AGH	-1.671	0.405	0.459	3.638
Total turnout	2023 AGH	-1.027	0.354	0.459	2.236
Total turnout	2024 European	-1.137	0.323	0.459	2.475
Total turnout	2025 federal	-0.806	0.419	0.459	1.754
Total turnout	2026 AGH	-0.836	0.323	0.459	1.820

Notes: Post-treatment coefficient magnitudes are compared with the largest absolute 2014/2016/2017 coefficient for each outcome. Ratios are descriptive, election-specific calibrations; they are not an HonestDiD identified set or a proof of parallel trends. The ratio compares each post-treatment coefficient to the largest observed pre-treatment coefficient in the same outcome; it does not model or bound post-treatment confounding.

Table B8: Precinct- and district-clustered uncertainty: In-person turnout

Election	Precinct cluster	District cluster
2014 EU	+0.326	+0.326
	(0.228)	(0.218)
2016 AGH	-0.228	-0.228
	(0.196)	(0.356)
2017 BTW	-0.226	-0.226
	(0.292)	(0.392)
2021 BTW	-1.078**	-1.078^{+}
	(0.373)	(0.504)
2021 AGH	-1.005**	-1.005^{+}
	(0.372)	(0.492)
2023 AGH	-1.145***	-1.145*
	(0.281)	(0.483)
2024 EU	-2.332***	-2.332***
	(0.247)	(0.411)
2025 BTW	-1.994***	-1.994*
	(0.415)	(0.656)
2026 AGH	-1.266***	-1.266*
	(0.346)	(0.517)
Observations	13,500	13,500
Precinct fixed effects	Yes	Yes
Election × district fixed effects	Yes	Yes

Notes: The same fixed-effects point estimates appear under both clustering schemes. Standard errors are clustered by 2021 postal precinct or by Berlin district (12 clusters); precincts are nested within districts, so two-way clustering by both dimensions is not an independent sensitivity. Significance markers follow the conventional clustered p-values.

Table B8: Precinct- and district-clustered uncertainty: Postal turnout (continued)

Election	Precinct cluster	District cluster
2014 EU	-0.251	-0.251
	(0.196)	(0.241)
2016 AGH	-0.231	-0.231
	(0.149)	(0.198)
2017 BTW	-0.058	-0.058
	(0.154)	(0.147)
2021 BTW	-0.547	-0.547
	(0.450)	(0.526)
2021 AGH	-0.666	-0.666
	(0.455)	(0.557)
2023 AGH	+0.118	+0.118
	(0.333)	(0.377)
2024 EU	+1.195***	+1.195**
	(0.298)	(0.319)
2025 BTW	+1.188***	+1.188*
	(0.336)	(0.437)
2026 AGH	+0.430	+0.430
	(0.310)	(0.307)
Observations	13,500	13,500
Precinct fixed effects	Yes	Yes
Election × district fixed effects	Yes	Yes

Notes: The same fixed-effects point estimates appear under both clustering schemes. Standard errors are clustered by 2021 postal precinct or by Berlin district (12 clusters); precincts are nested within districts, so two-way clustering by both dimensions is not an independent sensitivity. Significance markers follow the conventional clustered p-values.

Table B8: Precinct- and district-clustered uncertainty: Total turnout (continued)

Election	Precinct cluster	District cluster
2014 EU	+0.074	+0.074
	(0.269)	(0.275)
2016 AGH	-0.459^{+}	-0.459
	(0.270)	(0.528)
2017 BTW	-0.284	-0.284
	(0.301)	(0.400)
2021 BTW	-1.625***	-1.625**
	(0.393)	(0.395)
2021 AGH	-1.671***	-1.671**
	(0.405)	(0.438)
2023 AGH	-1.027**	-1.027^{+}
	(0.354)	(0.547)
2024 EU	-1.137***	-1.137**
	(0.323)	(0.365)
2025 BTW	-0.806^{+}	-0.806
	(0.419)	(0.524)
2026 AGH	-0.836**	-0.836^{+}
	(0.323)	(0.463)
Observations	13,500	13,500
Precinct fixed effects	Yes	Yes
Election × district fixed effects	Yes	Yes

Notes: The same fixed-effects point estimates appear under both clustering schemes. Standard errors are clustered by 2021 postal precinct or by Berlin district (12 clusters); precincts are nested within districts, so two-way clustering by both dimensions is not an independent sensitivity. Significance markers follow the conventional clustered p-values.

C Robustness of the participation response

The robustness checks target several separate threats. Within-election-type models examine whether the pooled estimates are sensitive to combining AGH, Bundestag, and European elections. Additional controls and matching address observable baseline differences; exclusion of district 07 and adjacent clean controls addresses particular reporting concerns and potential local contamination. Alternative geographic weights and the original polling-station-precinct analysis assess whether crosswalk construction drives the in-person pattern.

The resulting estimates are not interchangeable: sample restrictions change the population represented, and some election families contain different voting-eligibility rules. In-person turnout effects generally retain a negative sign across these comparisons, whereas net total-turnout effects are less precisely estimated in some reduced samples. The 2026 extension requires special caution because both the state franchise and the source-to-target geographic allocation differ from earlier years.

Table C1: Postal-turnout sensitivity to 2017–2025 geographic harmonization

Specification	2025 postal contrast	Observations	Precincts
Baseline	+1.246 (0.367)	2,700	1,350
HHI weighted	+1.245 (0.387)	2,700	1,350
k90 at most 2	+1.404 (0.394)	2,402	1,201

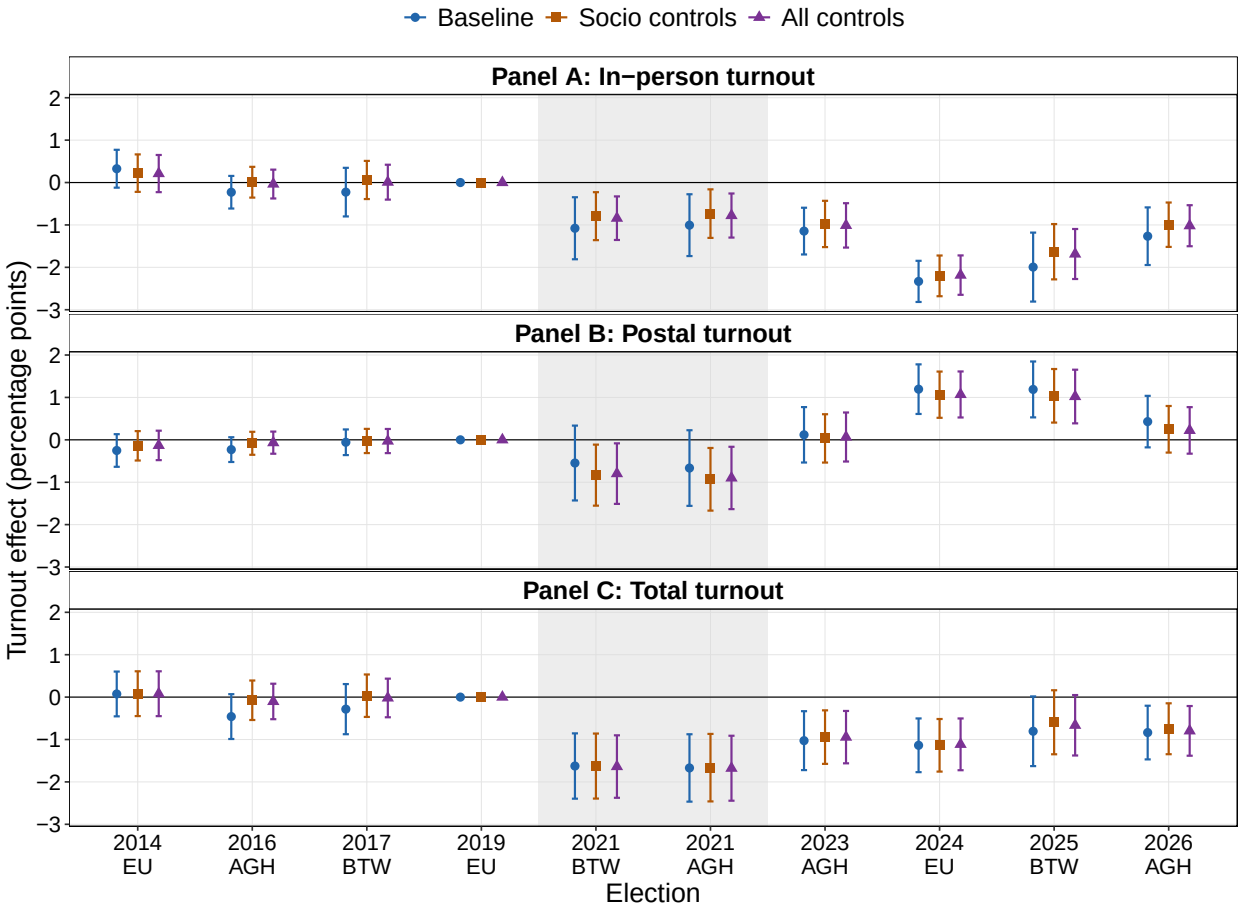
Notes: The 2017–2025 within-federal postal contrast is reported for the full sample, using crosswalk-HHI weights, and restricting to precincts whose postal allocation needs at most two source units to cover 90% of target mass in both years. The last restriction changes the analysis population. These comparisons use the existing 2017 and 2025 returns and source-to-target crosswalks.

Table C2: Persistence within election types: in-person, postal, and total turnout

Election	(1) In-person turnout	(2) Postal turnout	(3) Total turnout
Panel A: State elections (reference: 2016 AGH)			
2021 AGH	-0.777*	-0.435	-1.211**
	(0.307)	(0.477)	(0.395)
2023 AGH	-0.917***	0.349	-0.568
	(0.259)	(0.345)	(0.353)
2026 AGH	-1.037***	0.661+	-0.377
	(0.286)	(0.348)	(0.311)
Panel B: Federal elections (reference: 2017 federal)			
2021 federal	-0.852***	-0.490	-1.342**
	(0.221)	(0.460)	(0.430)
2025 federal	-1.768***	1.246***	-0.522
	(0.269)	(0.367)	(0.371)
Panel C: European elections (reference: 2019 European)			
2014 EU	0.326	-0.251	0.074
	(0.228)	(0.196)	(0.269)
2024 European	-2.332***	1.195***	-1.137***
	(0.247)	(0.298)	(0.323)
Fit statistics			
State elections observations	5,400	5,400	5,400
Federal elections observations	4,050	4,050	4,050
European elections observations	4,050	4,050	4,050
Fixed effects			
Precinct	Yes	Yes	Yes
Election × District	Yes	Yes	Yes

Notes: The table re-estimates the event-study specification within election families. These checks address the concern that the pooled panel combines federal, state, and European elections with different baseline turnout levels. State elections use 2016 AGH as their reference; European elections use 2019 EU, including the 2014 pre-breakdown comparison; 2026 has a newly eligible 16+ electorate. The table is descriptive of sensitivity to election-family comparisons, not a direct test of effects being equal across specifications.

Figure C1: Controls robustness: event-study estimates



Notes: The figure compares event-study coefficients across the baseline specification and models with additional controls. Estimates are shown for in-person, postal, and total turnout. The reference election is the 2019 European election. The companion table is Table C3.

Table C3: Additional-controls robustness: In-person turnout

Election	Baseline	Demographic controls	All controls
2014 EU	+0.326	+0.222	+0.212
	(0.228)	(0.225)	(0.223)
2016 AGH	-0.228	+0.008	-0.036
	(0.196)	(0.185)	(0.173)
2017 BTW	-0.226	+0.060	+0.009
	(0.292)	(0.229)	(0.210)
2021 BTW	-1.078**	-0.793**	-0.841**
	(0.373)	(0.288)	(0.262)
2021 AGH	-1.005**	-0.733*	-0.778**
	(0.372)	(0.292)	(0.265)
2023 AGH	-1.145***	-0.977***	-1.011***
	(0.281)	(0.278)	(0.267)
2024 EU	-2.332***	-2.201***	-2.184***
	(0.247)	(0.245)	(0.237)
2025 BTW	-1.994***	-1.632***	-1.686***
	(0.415)	(0.333)	(0.301)
2026 AGH	-1.266***	-0.995***	-1.018***
	(0.346)	(0.267)	(0.247)
Observations	13,500	13,500	13,500
Precinct fixed effects	Yes	Yes	Yes
Election × district fixed effects	Yes	Yes	Yes

Notes: Models retain 2021 postal-precinct and election-by-district fixed effects. Coefficients are changes relative to the 2019 European election; parenthesized standard errors are clustered by BWB21. Controls are measured before treatment. Source regressions and full estimates are unchanged.

Table C3: Additional-controls robustness: Postal turnout (continued)

Election	Baseline	Demographic controls	All controls
2014 EU	-0.251	-0.141	-0.132
	(0.196)	(0.177)	(0.177)
2016 AGH	-0.231	-0.082	-0.067
	(0.149)	(0.138)	(0.133)
2017 BTW	-0.058	-0.027	-0.029
	(0.154)	(0.146)	(0.145)
2021 BTW	-0.547	-0.833*	-0.797*
	(0.450)	(0.367)	(0.364)
2021 AGH	-0.666	-0.931*	-0.899*
	(0.455)	(0.377)	(0.375)
2023 AGH	+0.118	+0.034	+0.066
	(0.333)	(0.290)	(0.295)
2024 EU	+1.195***	+1.064***	+1.070***
	(0.298)	(0.278)	(0.277)
2025 BTW	+1.188***	+1.038**	+1.021**
	(0.336)	(0.322)	(0.322)
2026 AGH	+0.430	+0.248	+0.222
	(0.310)	(0.280)	(0.280)
Observations	13,500	13,500	13,500
Precinct fixed effects	Yes	Yes	Yes
Election × district fixed effects	Yes	Yes	Yes

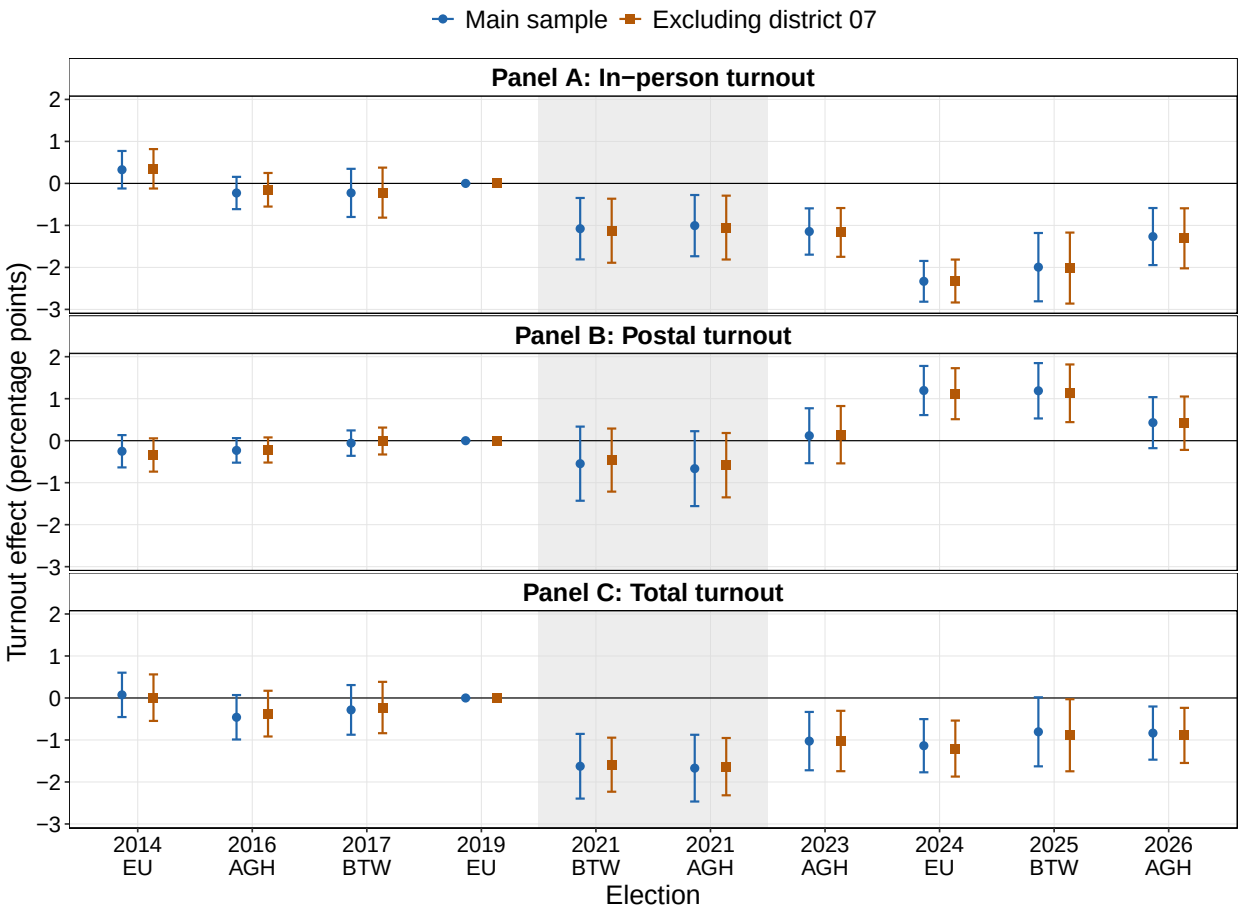
Notes: Models retain 2021 postal-precinct and election-by-district fixed effects. Coefficients are changes relative to the 2019 European election; parenthesized standard errors are clustered by BWB21. Controls are measured before treatment. Source regressions and full estimates are unchanged.

Table C3: Additional-controls robustness: Total turnout (continued)

Election	Baseline	Demographic controls	All controls
2014 EU	+0.074	+0.081	+0.080
	(0.269)	(0.269)	(0.270)
2016 AGH	-0.459^{+}	-0.074	-0.103
	(0.270)	(0.237)	(0.213)
2017 BTW	-0.284	+0.033	-0.020
	(0.301)	(0.255)	(0.232)
2021 BTW	-1.625***	-1.626***	-1.638***
	(0.393)	(0.391)	(0.376)
2021 AGH	-1.671***	-1.664***	-1.678***
	(0.405)	(0.406)	(0.391)
2023 AGH	-1.027**	-0.943**	-0.944**
	(0.354)	(0.322)	(0.315)
2024 EU	-1.137***	-1.137***	-1.114***
	(0.323)	(0.316)	(0.311)
2025 BTW	-0.806^{+}	-0.593	-0.664^{+}
	(0.419)	(0.385)	(0.363)
2026 AGH	-0.836**	-0.746*	-0.796**
	(0.323)	(0.306)	(0.299)
Observations	13,500	13,500	13,500
Precinct fixed effects	Yes	Yes	Yes
Election × district fixed effects	Yes	Yes	Yes

Notes: Models retain 2021 postal-precinct and election-by-district fixed effects. Coefficients are changes relative to the 2019 European election; parenthesized standard errors are clustered by BWB21. Controls are measured before treatment. Source regressions and full estimates are unchanged.

Figure C2: Event-study estimates excluding district 07



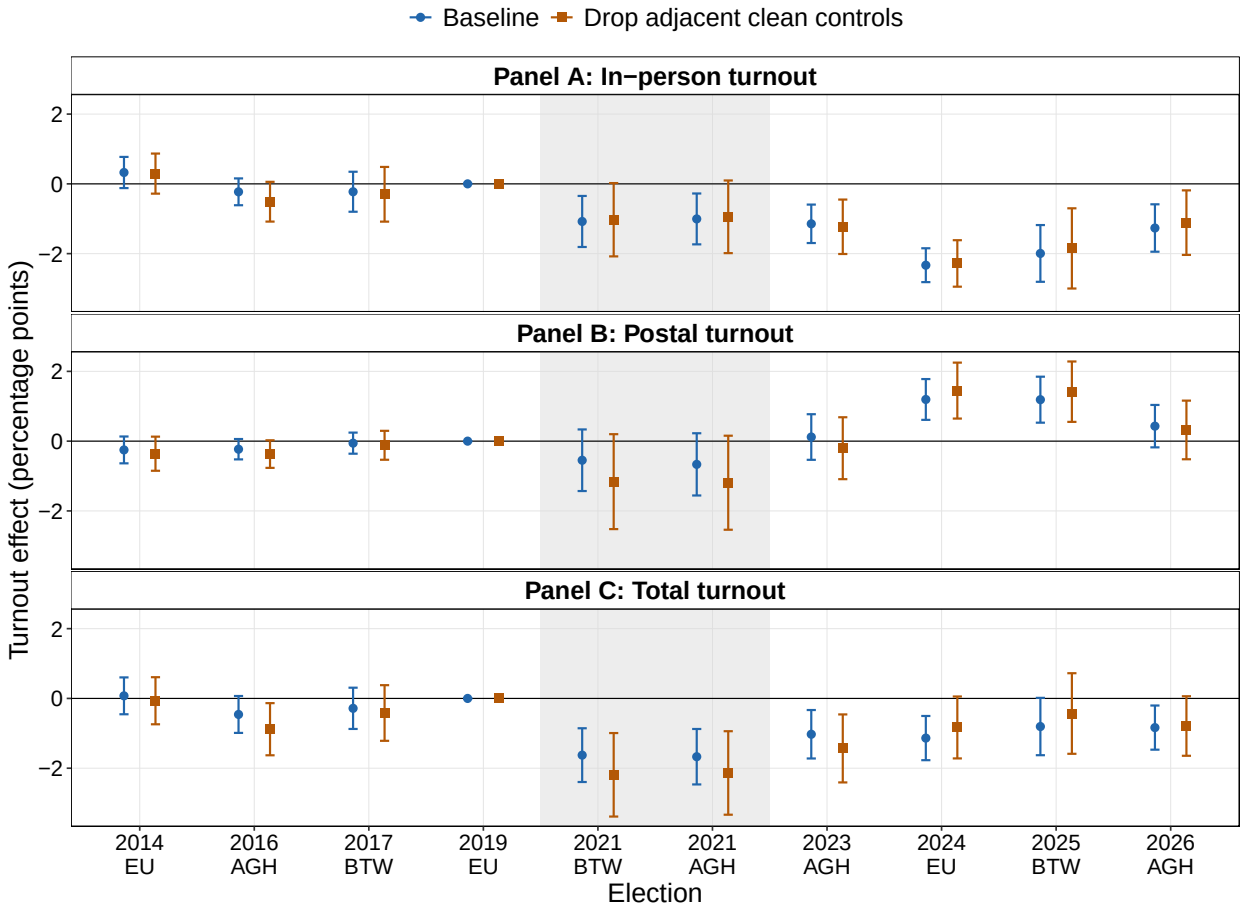
Notes: The figure plots the main event-study estimates after excluding district 07. This check addresses the fact that allocation to 2021 postal precincts was not unambiguous for the 2021 election in district 07. Outcomes are turnout levels in percentage points. The companion table is Table C4.

Table C4: Robustness excluding district 07

Election	(1) In-person turnout	(2) Postal turnout	(3) Total turnout
2014 EU	0.348	-0.340^{+}	0.007
	(0.239)	(0.202)	(0.282)
2016 AGH	-0.151	-0.221	-0.373
	(0.204)	(0.152)	(0.278)
2017 federal	-0.220	-0.008	-0.227
	(0.303)	(0.163)	(0.312)
2021 federal	-1.127**	-0.462	-1.589***
	(0.389)	(0.383)	(0.328)
2021 AGH	-1.052**	-0.583	-1.635***
	(0.387)	(0.391)	(0.348)
2023 AGH	-1.167***	0.143	-1.024**
	(0.296)	(0.348)	(0.366)
2024 European	-2.324***	1.119***	-1.205***
	(0.260)	(0.310)	(0.340)
2025 federal	-2.016***	1.129**	-0.887*
	(0.431)	(0.350)	(0.437)
2026 AGH	-1.307***	0.416	-0.891**
	(0.364)	(0.324)	(0.334)
Fit statistics			
Observations	12,260	12,260	12,260
Fixed effects			
Precinct	Yes	Yes	Yes
Election × District	Yes	Yes	Yes

Notes: The table replicates the main event-study specification after excluding district 07, where allocation to 2021 postal precincts was not unambiguous for the 2021 election. The corresponding figure is Figure C2.

Figure C3: Adjacent-control spillover check



Notes: The figure compares the baseline estimates to estimates dropping clean-control precincts adjacent to treated precincts. This is a conservative local-spillover diagnostic. The reference election is the 2019 European election. The companion table is Table C5.

Table C5: Adjacent-control restriction: In-person turnout

Election	Baseline	Exclude adjacent controls
2014 EU	+0.326	+0.294
	(0.228)	(0.293)
2016 AGH	-0.228	-0.512^{+}
	(0.196)	(0.290)
2017 BTW	-0.226	-0.299
	(0.292)	(0.399)
2021 BTW	-1.078**	-1.028^{+}
	(0.373)	(0.535)
2021 AGH	-1.005**	-0.944^{+}
	(0.372)	(0.531)
2023 AGH	-1.145***	-1.230**
	(0.281)	(0.398)
2024 EU	-2.332***	-2.281***
	(0.247)	(0.339)
2025 BTW	-1.994***	-1.850**
	(0.415)	(0.586)
2026 AGH	-1.266***	-1.112*
	(0.346)	(0.471)
Observations	13,500	8,950
Precinct fixed effects	Yes	Yes
Election × district fixed effects	Yes	Yes

Notes: The restricted model excludes clean-control BWB21 areas adjacent to court-linked areas. The 2019 European election is the reference; standard errors are clustered by 2021 postal precinct. The comparison changes the control sample and is not an independent estimate of a spillover effect.

Table C5: Adjacent-control restriction: Postal turnout (continued)

Election	Baseline	Exclude adjacent controls
2014 EU	-0.251	-0.360
	(0.196)	(0.249)
2016 AGH	-0.231	-0.371^{+}
	(0.149)	(0.201)
2017 BTW	-0.058	-0.119
	(0.154)	(0.211)
2021 BTW	-0.547	-1.161^{+}
	(0.450)	(0.693)
2021 AGH	-0.666	-1.192^{+}
	(0.455)	(0.687)
2023 AGH	+0.118	-0.203
	(0.333)	(0.452)
2024 EU	+1.195***	+1.449***
	(0.298)	(0.408)
2025 BTW	+1.188***	+1.418**
	(0.336)	(0.441)
2026 AGH	+0.430	+0.321
	(0.310)	(0.428)
Observations	13,500	8,950
Precinct fixed effects	Yes	Yes
Election × district fixed effects	Yes	Yes

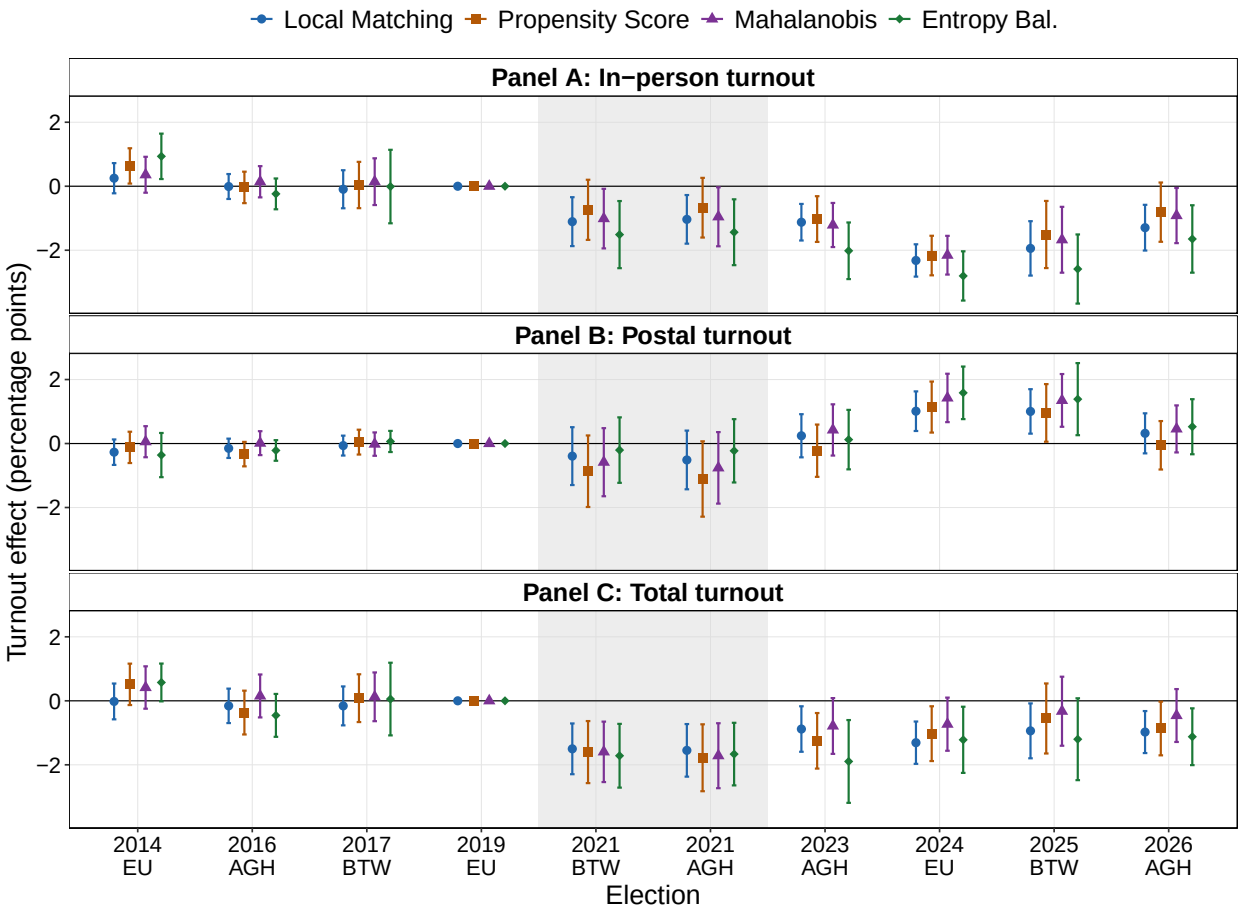
Notes: The restricted model excludes clean-control BWB21 areas adjacent to court-linked areas. The 2019 European election is the reference; standard errors are clustered by 2021 postal precinct. The comparison changes the control sample and is not an independent estimate of a spillover effect.

Table C5: Adjacent-control restriction: Total turnout (continued)

Election	Baseline	Exclude adjacent controls
2014 EU	+0.074	-0.066
	(0.269)	(0.344)
2016 AGH	-0.459^{+}	-0.882*
	(0.270)	(0.381)
2017 BTW	-0.284	-0.418
	(0.301)	(0.406)
2021 BTW	-1.625***	-2.189***
	(0.393)	(0.610)
2021 AGH	-1.671***	-2.136***
	(0.405)	(0.610)
2023 AGH	-1.027**	-1.433**
	(0.354)	(0.497)
2024 EU	-1.137***	-0.832^{+}
	(0.323)	(0.452)
2025 BTW	-0.806^{+}	-0.432
	(0.419)	(0.589)
2026 AGH	-0.836**	-0.791^{+}
	(0.323)	(0.435)
Observations	13,500	8,950
Precinct fixed effects	Yes	Yes
Election × district fixed effects	Yes	Yes

Notes: The restricted model excludes clean-control BWB21 areas adjacent to court-linked areas. The 2019 European election is the reference; standard errors are clustered by 2021 postal precinct. The comparison changes the control sample and is not an independent estimate of a spillover effect.

Figure C4: Matching robustness by turnout mode



Notes: The figure reports event-study estimates for in-person, postal, and total turnout under four matching specifications: local matching, propensity-score matching, Mahalanobis matching, and entropy balancing. The reference election is the 2019 European election. The companion table is Table C6.

Table C6: Matching robustness: In-person turnout

Election	Local	Propensity score	Mahalanobis	Entropy balance
2014 EU	+0.251	+0.635*	+0.359	+0.934**
	(0.240)	(0.281)	(0.286)	(0.362)
2016 AGH	-0.008	-0.037	+0.140	-0.239
	(0.198)	(0.250)	(0.248)	(0.245)
2017 BTW	-0.094	+0.039	+0.143	-0.010
	(0.303)	(0.368)	(0.372)	(0.585)
2021 BTW	-1.107**	-0.737	-1.012*	-1.512**
	(0.389)	(0.478)	(0.474)	(0.535)
2021 AGH	-1.035**	-0.671	-0.955*	-1.438**
	(0.387)	(0.474)	(0.469)	(0.525)
2023 AGH	-1.124***	-1.026**	-1.212***	-2.017***
	(0.291)	(0.364)	(0.351)	(0.451)
2024 EU	-2.320***	-2.166***	-2.155***	-2.803***
	(0.257)	(0.315)	(0.308)	(0.392)
2025 BTW	-1.942***	-1.509**	-1.673**	-2.587***
	(0.433)	(0.534)	(0.524)	(0.551)
2026 AGH	-1.295***	-0.810^{+}	-0.916*	-1.648**
	(0.364)	(0.471)	(0.439)	(0.537)
Observations	7,290	4,540	4,560	13,500
Precinct fixed effects	Yes	Yes	Yes	Yes
Election × district fixed effects	Yes	Yes	Yes	Yes

Notes: Each column uses its stated matching or balancing specification; the underlying sample size may differ across columns. The 2019 European election is omitted; parenthesized standard errors are clustered by the 2021 postal precinct. Covariate-balance diagnostics are reported in the replication diagnostics.

Table C6: Matching robustness: Postal turnout (continued)

Election	Local	Propensity score	Mahalanobis	Entropy balance
2014 EU	-0.271	-0.121	+0.057	-0.361
	(0.203)	(0.249)	(0.247)	(0.352)
2016 AGH	-0.149	-0.331^{+}	+0.011	-0.217
	(0.154)	(0.195)	(0.190)	(0.164)
2017 BTW	-0.065	+0.044	-0.018	+0.064
	(0.158)	(0.197)	(0.186)	(0.168)
2021 BTW	-0.394	-0.866	-0.584	-0.205
	(0.461)	(0.569)	(0.542)	(0.522)
2021 AGH	-0.513	-1.109^{+}	-0.760	-0.228
	(0.467)	(0.599)	(0.569)	(0.504)
2023 AGH	+0.242	-0.224	+0.425	+0.122
	(0.343)	(0.416)	(0.407)	(0.474)
2024 EU	+1.011**	+1.139**	+1.425***	+1.584***
	(0.315)	(0.406)	(0.385)	(0.419)
2025 BTW	+1.003**	+0.957*	+1.347**	+1.387*
	(0.354)	(0.458)	(0.419)	(0.574)
2026 AGH	+0.318	-0.055	+0.457	+0.525
	(0.319)	(0.385)	(0.374)	(0.439)
Observations	7,290	4,540	4,560	13,500
Precinct fixed effects	Yes	Yes	Yes	Yes
Election × district fixed effects	Yes	Yes	Yes	Yes

Notes: Each column uses its stated matching or balancing specification; the underlying sample size may differ across columns. The 2019 European election is omitted; parenthesized standard errors are clustered by the 2021 postal precinct. Covariate-balance diagnostics are reported in the replication diagnostics.

Table C6: Matching robustness: Total turnout (continued)

Election	Local	Propensity score	Mahalanobis	Entropy balance
2014 EU	-0.020	+0.514	+0.416	+0.573^{+}
	(0.285)	(0.329)	(0.337)	(0.301)
2016 AGH	-0.158	-0.368	+0.152	-0.455
	(0.274)	(0.348)	(0.341)	(0.341)
2017 BTW	-0.159	+0.082	+0.125	+0.055
	(0.310)	(0.380)	(0.387)	(0.579)
2021 BTW	-1.501***	-1.603**	-1.595***	-1.717***
	(0.404)	(0.493)	(0.481)	(0.508)
2021 AGH	-1.549***	-1.780***	-1.716***	-1.665***
	(0.419)	(0.532)	(0.517)	(0.499)
2023 AGH	-0.882*	-1.250**	-0.787^{+}	-1.895**
	(0.362)	(0.442)	(0.443)	(0.659)
2024 EU	-1.309***	-1.027*	-0.731^{+}	-1.219*
	(0.337)	(0.436)	(0.423)	(0.527)
2025 BTW	-0.939*	-0.552	-0.325	-1.200^{+}
	(0.437)	(0.557)	(0.549)	(0.652)
2026 AGH	-0.977**	-0.865*	-0.460	-1.123*
	(0.334)	(0.427)	(0.421)	(0.453)
Observations	7,290	4,540	4,560	13,500
Precinct fixed effects	Yes	Yes	Yes	Yes
Election × district fixed effects	Yes	Yes	Yes	Yes

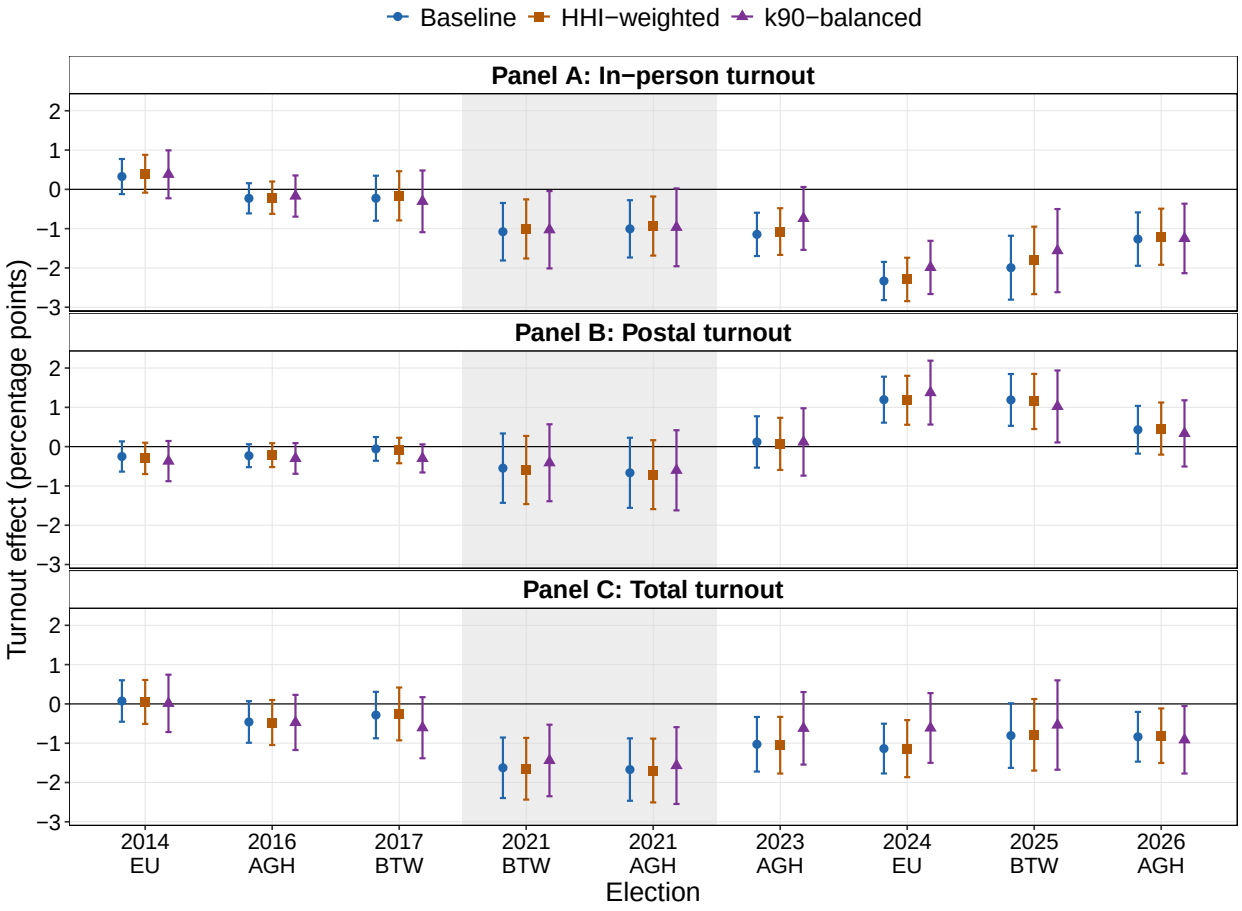
Notes: Each column uses its stated matching or balancing specification; the underlying sample size may differ across columns. The 2019 European election is omitted; parenthesized standard errors are clustered by the 2021 postal precinct. Covariate-balance diagnostics are reported in the replication diagnostics.

Table C7: Matching robustness: retained estimation observations

Matching approach	Treated units	Control units	In-person N	Postal N	Total N
Local Matching	294	435	7290	7290	7290
Propensity Score	294	160	4540	4540	4540
Mahalanobis	294	162	4560	4560	4560
Entropy Bal.	294	1056	13500	13500	13500

Notes: Each method changes the effective estimation sample or weighting. The counts are reported separately by voting mode because missing outcomes can change N. Local matching, propensity-score matching, Mahalanobis matching, and entropy balancing target the treated–control contrast under distinct balance/overlap choices; they do not guarantee common support across all covariates. These counts document retention, not a proof of balance.

Figure C5: Harmonization robustness: concentration weighting and full-panel k90 restriction



Notes: Historical precinct boundaries are mapped to the 2021 postal-precinct geography with population-weighted crosswalks. The baseline specification uses the full harmonized panel. The HHI-weighted specification assigns more weight to observations whose source-to-target mapping is more concentrated. The full-panel k90 specification restricts the sample to harmonized units for which a small number of source units accounts for most of the 2021 target mass. The reference election is the 2019 European election. The companion table is Table C8.

Table C8: Spatial-harmonization robustness: In-person turnout

Election	Baseline	HHI weighted	Full-panel k90
2014 EU	+0.326	+0.397	+0.382
	(0.228)	(0.246)	(0.311)
2016 AGH	-0.228	-0.213	-0.171
	(0.196)	(0.210)	(0.267)
2017 BTW	-0.226	-0.164	-0.305
	(0.292)	(0.319)	(0.400)
2021 BTW	-1.078**	-1.007**	-1.028*
	(0.373)	(0.384)	(0.501)
2021 AGH	-1.005**	-0.934*	-0.966+
	(0.372)	(0.383)	(0.504)
2023 AGH	-1.145***	-1.074***	-0.740+
	(0.281)	(0.303)	(0.408)
2024 EU	-2.332***	-2.293***	-1.987***
	(0.247)	(0.282)	(0.345)
2025 BTW	-1.994***	-1.808***	-1.560**
	(0.415)	(0.438)	(0.539)
2026 AGH	-1.266***	-1.206***	-1.249**
	(0.346)	(0.364)	(0.450)
Observations	13,500	13,500	7,050
Precinct fixed effects	Yes	Yes	Yes
Election × district fixed effects	Yes	Yes	Yes

Notes: HHI denotes the Herfindahl concentration index of source-to-target crosswalk weights. Full-panel k90 retains units satisfying the crosswalk-concentration restriction across the relevant panel elections; it is not the federal-pair k90 sample in Table C1. The 2019 European election is the reference; parentheses contain BWB21-clustered standard errors.

Table C8: Spatial-harmonization robustness: Postal turnout (continued)

Election	Baseline	HHI weighted	Full-panel k90
2014 EU	-0.251	-0.299	-0.368
	(0.196)	(0.203)	(0.261)
2016 AGH	-0.231	-0.217	-0.301
	(0.149)	(0.155)	(0.199)
2017 BTW	-0.058	-0.098	-0.300^{+}
	(0.154)	(0.165)	(0.182)
2021 BTW	-0.547	-0.595	-0.411
	(0.450)	(0.442)	(0.499)
2021 AGH	-0.666	-0.713	-0.602
	(0.455)	(0.447)	(0.519)
2023 AGH	+0.118	+0.071	+0.119
	(0.333)	(0.338)	(0.437)
2024 EU	+1.195***	+1.179***	+1.375***
	(0.298)	(0.318)	(0.414)
2025 BTW	+1.188***	+1.150**	+1.023*
	(0.336)	(0.358)	(0.466)
2026 AGH	+0.430	+0.459	+0.337
	(0.310)	(0.338)	(0.429)
Observations	13,500	13,500	7,050
Precinct fixed effects	Yes	Yes	Yes
Election × district fixed effects	Yes	Yes	Yes

Notes: HHI denotes the Herfindahl concentration index of source-to-target crosswalk weights. Full-panel k90 retains units satisfying the crosswalk-concentration restriction across the relevant panel elections; it is not the federal-pair k90 sample in Table C1. The 2019 European election is the reference; parentheses contain BWB21-clustered standard errors.

Table C8: Spatial-harmonization robustness: Total turnout (continued)

Election	Baseline	HHI weighted	Full-panel k90
2014 EU	+0.074	+0.050	+0.014
	(0.269)	(0.285)	(0.372)
2016 AGH	-0.459^{+}	-0.473	-0.471
	(0.270)	(0.292)	(0.357)
2017 BTW	-0.284	-0.255	-0.606
	(0.301)	(0.343)	(0.396)
2021 BTW	-1.625***	-1.650***	-1.439**
	(0.393)	(0.400)	(0.464)
2021 AGH	-1.671***	-1.695***	-1.569**
	(0.405)	(0.414)	(0.498)
2023 AGH	-1.027**	-1.051**	-0.621
	(0.354)	(0.368)	(0.470)
2024 EU	-1.137***	-1.138**	-0.612
	(0.323)	(0.370)	(0.452)
2025 BTW	-0.806^{+}	-0.786^{+}	-0.537
	(0.419)	(0.464)	(0.579)
2026 AGH	-0.836**	-0.809*	-0.912*
	(0.323)	(0.354)	(0.437)
Observations	13,500	13,500	7,050
Precinct fixed effects	Yes	Yes	Yes
Election × district fixed effects	Yes	Yes	Yes

Notes: HHI denotes the Herfindahl concentration index of source-to-target crosswalk weights. Full-panel k90 retains units satisfying the crosswalk-concentration restriction across the relevant panel elections; it is not the federal-pair k90 sample in Table C1. The 2019 European election is the reference; parentheses contain BWB21-clustered standard errors.

Table C9: Crosswalk restriction: characteristics of retained and omitted units

Sample	BWB21 areas	Treated share	Mean residents	Mean 2019 in-person turnout
Not retained by k90	645	0.239	1736.690	41.918
Retained by k90	705	0.199	1656.651	43.189

Notes: Baseline characteristics are measured in the 2019 EU election on 2021 postal-precinct geography. The k90 retention indicator is drawn from the same restricted-crosswalk sample used in the robustness regressions. Group differences are descriptive and do not imply that the retained sample represents the full population.

Table C10: Auxiliary polling-station-precinct analysis for in-person turnout

Election	(1) In-person turnout
2014 EU	0.308
	(0.217)
2016 AGH	-0.262
	(0.182)
2017 federal	-0.318
	(0.256)
2021 federal	-1.212***
	(0.337)
2021 AGH	-1.117***
	(0.334)
2023 AGH	-1.254***
	(0.253)
2024 European	-2.430***
	(0.233)
2025 federal	-2.267***
	(0.378)
2026 AGH	-1.439***
	(0.319)
Fit statistics	
Observations	20,958
Fixed effects	
Polling-station precinct	Yes
Election × District	Yes

Notes: The table reports in-person turnout models at the polling-station-precinct geography. The analysis is auxiliary because postal and total turnout are analyzed at the postal-precinct geography in the main panel. The corresponding geographic comparison appears in main-text Figure 3.

Table C11: Turnout-above-100 exclusion sensitivity: 2026 state-election estimates

Outcome	Full analysis sample	Excluding excess-turnout rows
In-person turnout	-1.266 (0.346)	-1.266 (0.346)
Postal turnout	+0.430 (0.310)	+0.430 (0.310)
Total turnout	-0.836 (0.323)	-0.836 (0.323)

Notes: Entries are event-study coefficients in percentage points, with precinct-clustered standard errors in parentheses and the 2019 European election as reference. The restricted sample removes all 18 precinct-election observations whose interpolated total turnout exceeds 100 percent (17 in the two 2021 returns and one in the 2025 federal return; 16 lie in district 07). The reported comparison concerns the terminal 2026 contrast and does not establish an unchanged full event-study path. The R output also saves event-by-event estimates for verification after re-estimation.

Table C12: Omission and reference-election checks: In-person turnout

Election	Omit 2023 AGH	Omit 2024 EU	Post-2023 only	Before 2024 rerun
2014 EU	+0.326	+0.326	—	+0.326
	(0.228)	(0.228)		(0.228)
2016 AGH	-0.228	-0.228	—	-0.228
	(0.196)	(0.196)		(0.196)
2017 BTW	-0.226	-0.226	—	-0.226
	(0.292)	(0.292)		(0.292)
2021 BTW	-1.078**	-1.078**	—	-1.078**
	(0.373)	(0.373)		(0.373)
2021 AGH	-1.005**	-1.005**	—	-1.005**
	(0.372)	(0.372)		(0.372)
2023 AGH	—	-1.145***	—	-1.145***
		(0.281)		(0.281)
2024 EU	-2.332***	—	-1.186***	—
	(0.247)		(0.184)	
2025 BTW	-1.994***	-1.994***	-0.849**	—
	(0.415)	(0.415)	(0.290)	
2026 AGH	-1.266***	-1.266***	-0.120	—
	(0.346)	(0.346)	(0.269)	
Observations	12,150	12,150	5,400	9,450
Precinct fixed effects	Yes	Yes	Yes	Yes
Election × district fixed effects	Yes	Yes	Yes	Yes

Notes: Columns omit the named election(s) or use the stated time window. The post-2023 specification uses the 2023 AGH reference; the other columns retain 2019 EU. An em dash denotes an omitted or reference election rather than a zero coefficient. Parentheses contain BWB21-clustered standard errors.

Table C12: Omission and reference-election checks: Postal turnout (continued)

Election	Omit 2023 AGH	Omit 2024 EU	Post-2023 only	Before 2024 rerun
2014 EU	-0.251 (0.196)	-0.251 (0.196)	—	-0.251 (0.196)
2016 AGH	-0.231 (0.149)	-0.231 (0.149)	—	-0.231 (0.149)
2017 BTW	-0.058 (0.154)	-0.058 (0.154)	—	-0.058 (0.154)
2021 BTW	-0.547 (0.450)	-0.547 (0.450)	—	-0.547 (0.450)
2021 AGH	-0.666 (0.455)	-0.666 (0.455)	—	-0.666 (0.455)
2023 AGH	—	+0.118 (0.333)	—	+0.118 (0.333)
2024 EU	+1.195*** (0.298)	—	+1.077*** (0.303)	—
2025 BTW	+1.188*** (0.336)	+1.188*** (0.336)	+1.070** (0.334)	—
2026 AGH	+0.430 (0.310)	+0.430 (0.310)	+0.311 (0.256)	—
Observations	12,150	12,150	5,400	9,450
Precinct fixed effects	Yes	Yes	Yes	Yes
Election × district fixed effects	Yes	Yes	Yes	Yes

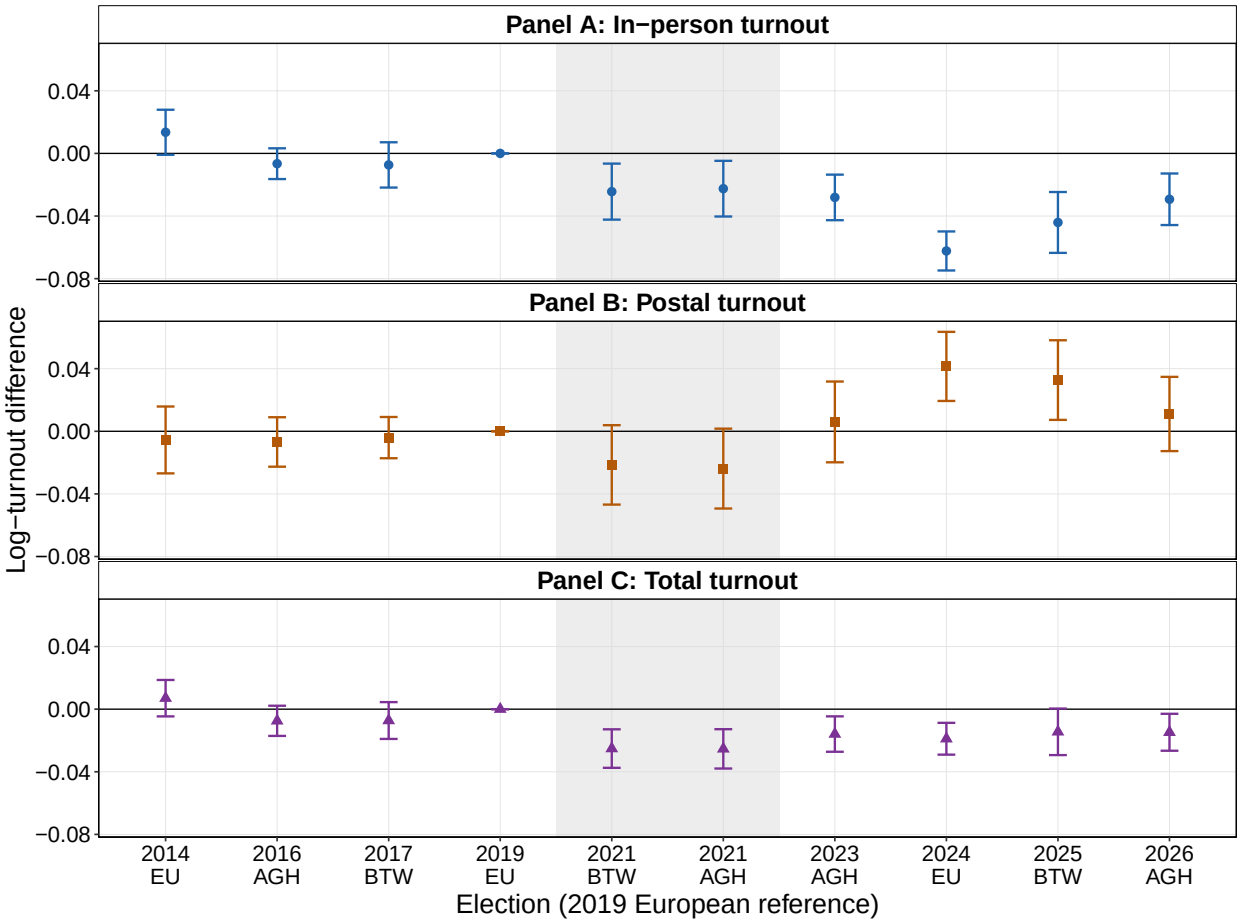
Notes: Columns omit the named election(s) or use the stated time window. The post-2023 specification uses the 2023 AGH reference; the other columns retain 2019 EU. An em dash denotes an omitted or reference election rather than a zero coefficient. Parentheses contain BWB21-clustered standard errors.

Table C12: Omission and reference-election checks: Total turnout (continued)

Election	Omit 2023 AGH	Omit 2024 EU	Post-2023 only	Before 2024 rerun
2014 EU	+0.074	+0.074	—	+0.074
	(0.269)	(0.269)		(0.269)
2016 AGH	-0.459^{+}	-0.459^{+}	—	-0.459^{+}
	(0.270)	(0.270)		(0.270)
2017 BTW	-0.284	-0.284	—	-0.284
	(0.301)	(0.301)		(0.301)
2021 BTW	-1.625***	-1.625***	—	-1.625***
	(0.393)	(0.393)		(0.393)
2021 AGH	-1.671***	-1.671***	—	-1.671***
	(0.405)	(0.405)		(0.405)
2023 AGH	—	-1.027**	—	-1.027**
		(0.354)		(0.354)
2024 EU	-1.137***	—	-0.110	—
	(0.323)		(0.328)	
2025 BTW	-0.806^{+}	-0.806^{+}	+0.221	—
	(0.419)	(0.419)	(0.424)	
2026 AGH	-0.836**	-0.836**	+0.191	—
	(0.323)	(0.323)	(0.329)	
Observations	12,150	12,150	5,400	9,450
Precinct fixed effects	Yes	Yes	Yes	Yes
Election × district fixed effects	Yes	Yes	Yes	Yes

Notes: Columns omit the named election(s) or use the stated time window. The post-2023 specification uses the 2023 AGH reference; the other columns retain 2019 EU. An em dash denotes an omitted or reference election rather than a zero coefficient. Parentheses contain BWB21-clustered standard errors.

Figure C6: Log-turnout sensitivity



Notes: The figure replicates the main event-study specification using logged turnout rates. The preferred specification in the main text uses turnout levels in percentage points because the substantive estimand is an absolute participation change. The companion table is Table C13.

Table C13: Log-turnout sensitivity

Election	(1) In-person turnout	(2) Postal turnout	(3) Total turnout
2014 EU	0.014^{+}	-0.006	0.007
	(0.007)	(0.011)	(0.006)
2016 AGH	-0.007	-0.007	-0.007
	(0.005)	(0.008)	(0.005)
2017 federal	-0.007	-0.004	-0.007
	(0.007)	(0.007)	(0.006)
2021 federal	-0.024**	-0.021^{+}	-0.025***
	(0.009)	(0.013)	(0.006)
2021 AGH	-0.023*	-0.024^{+}	-0.025***
	(0.009)	(0.013)	(0.006)
2023 AGH	-0.028***	0.006	-0.016**
	(0.007)	(0.013)	(0.006)
2024 European	-0.062***	0.041***	-0.019***
	(0.006)	(0.011)	(0.005)
2025 federal	-0.044***	0.033*	-0.014^{+}
	(0.010)	(0.013)	(0.008)
2026 AGH	-0.029***	0.011	-0.015*
	(0.008)	(0.012)	(0.006)
Fit statistics			
Observations	13,500	13,500	13,500
Fixed effects			
Precinct	Yes	Yes	Yes
Election × District	Yes	Yes	Yes

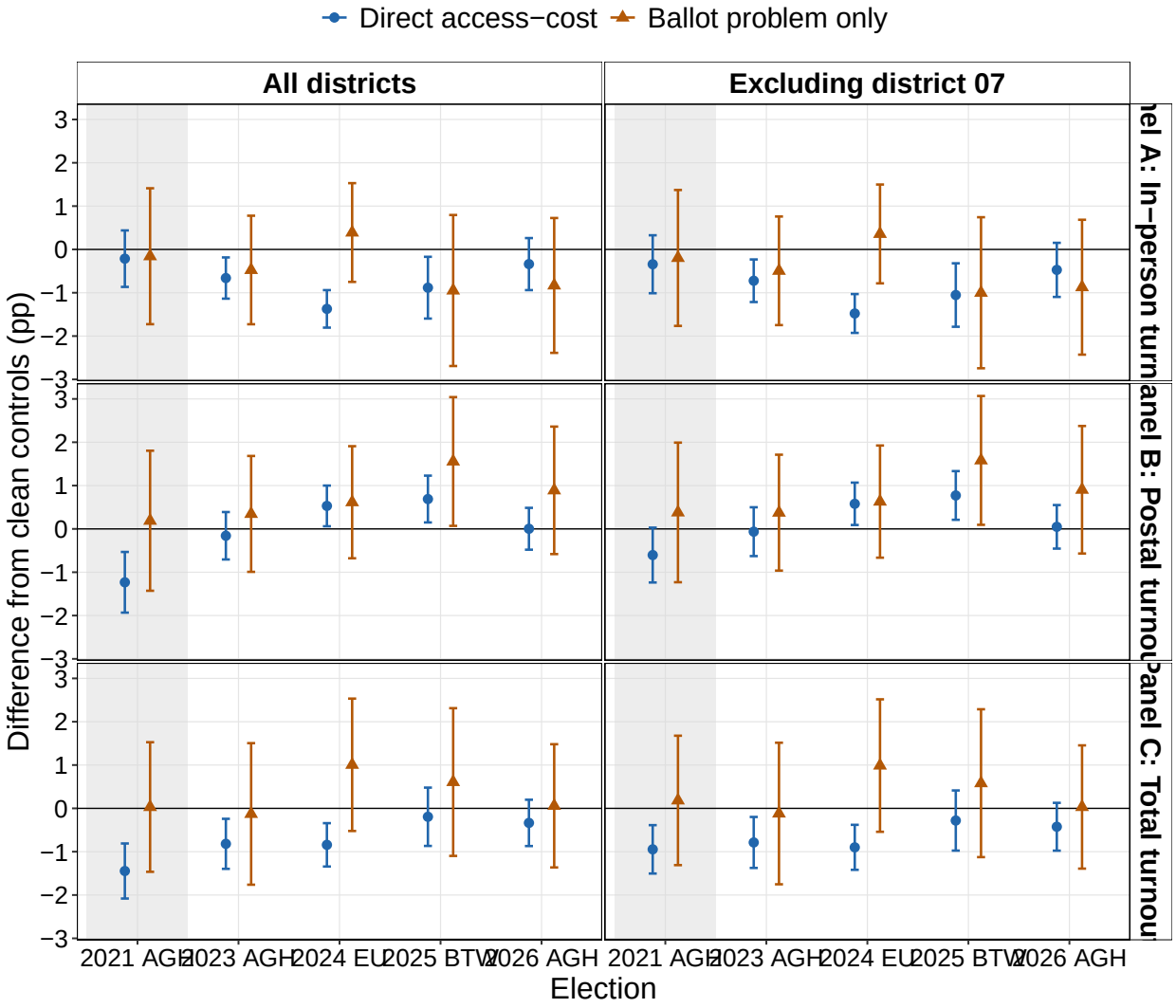
Notes: The table replicates the main event-study specification using logged turnout rates. The preferred specification uses turnout levels in percentage points because the substantive estimand is an absolute participation change. The corresponding figure is Figure C6.

D Disruption families and severity

The preferred treatment definition remains the federal court-linked record. State records permit a supplementary distinction between direct impediments to polling-place access and ballot errors without a recorded access incident. These groups differ in classification, coverage, and pre-treatment characteristics. The pooled comparisons below estimate both families against the same clean controls; they do not identify an experimentally assigned mechanism.

The strongest between-family contrast occurs for in-person turnout in the 2024 European election. Other election-specific comparisons are less precisely estimated, and the direct-access group's contemporaneous 2021 postal results cannot be interpreted as switching in response to later election-day problems. Excluding district 07 addresses a documented postal-allocation concern; it does not generally correct all election returns. Figure D1 and Table D1 retain the same-sample family comparison and its uncertainty. The estimation scripts also generate full event-by-family coefficients; the appendix emphasizes the main between-family contrasts.

Figure D1: Within-family turnout responses: all districts and excluding Tempelhof-Schöneberg



Notes: Separate event-study paths for the access-cost and ballot-only families are estimated in a single pooled model with a common clean-control group. The left panels include Berlin; the right panels exclude district 07. The reference is the 2019 European election; intervals are 95% cluster-robust confidence intervals. The 2021 postal estimates are contemporaneous diagnostics, not evidence of switching caused by election-day disruption. The accompanying numerical estimates and pooled contrasts are reported below.

Table D1: Disruption-family differences: In-person turnout

Election	All districts	Excluding district 07
2021 AGH	-0.056	-0.146
	(0.827)	(0.828)
2023 AGH	-0.186	-0.229
	(0.658)	(0.659)
2024 EU	-1.763	-1.836
	(0.604)	(0.606)
2025 BTW	+0.065	-0.053
	(0.916)	(0.917)
2026 AGH	+0.493	+0.400
	(0.817)	(0.819)
Joint test: 2023–2026	$p < 0.001$	$p < 0.001$
Observations	14,900	13,610

Notes: Each estimate is direct access-cost minus ballot-problem-only, obtained in a common pooled model with identical clean-control areas. Parentheses contain BWB21-clustered standard errors; the joint test evaluates the four election-specific differences for 2023 AGH, 2024 EU, 2025 BTW and 2026 AGH. In-person, postal and total turnout are estimated separately. The 2021 postal estimate is contemporaneous evidence, not identified mode substitution.

Table D1: Disruption-family differences: Postal turnout (continued)

Election	All districts	Excluding district 07
2021 AGH	-1.421	-0.984
	(0.854)	(0.845)
2023 AGH	-0.504	-0.439
	(0.707)	(0.709)
2024 EU	-0.085	-0.051
	(0.666)	(0.667)
2025 BTW	-0.867	-0.809
	(0.778)	(0.779)
2026 AGH	-0.887	-0.856
	(0.766)	(0.768)
Joint test: 2023–2026	$p = 0.083$	$p = 0.101$
Observations	14,900	13,610

Notes: Each estimate is direct access-cost minus ballot-problem-only, obtained in a common pooled model with identical clean-control areas. Parentheses contain BWB21-clustered standard errors; the joint test evaluates the four election-specific differences for 2023 AGH, 2024 EU, 2025 BTW and 2026 AGH. In-person, postal and total turnout are estimated separately. The 2021 postal estimate is contemporaneous evidence, not identified mode substitution.

Table D1: Disruption-family differences: Total turnout (continued)

Election	All districts	Excluding district 07
2021 AGH	-1.477	-1.129
	(0.802)	(0.795)
2023 AGH	-0.690	-0.668
	(0.863)	(0.865)
2024 EU	-1.848	-1.888
	(0.788)	(0.789)
2025 BTW	-0.802	-0.862
	(0.887)	(0.888)
2026 AGH	-0.394	-0.456
	(0.746)	(0.747)
Joint test: 2023–2026	$p = 0.138$	$p = 0.124$
Observations	14,900	13,610

Notes: Each estimate is direct access-cost minus ballot-problem-only, obtained in a common pooled model with identical clean-control areas. Parentheses contain BWB21-clustered standard errors; the joint test evaluates the four election-specific differences for 2023 AGH, 2024 EU, 2025 BTW and 2026 AGH. In-person, postal and total turnout are estimated separately. The 2021 postal estimate is contemporaneous evidence, not identified mode substitution.

Waiting/closure severity offers a different, more exploratory exposure measure. Its pre-2021 coefficients are appreciably larger than the preferred binary treatment's pre-period estimates, which limits any causal dose interpretation. Table D2 retains the relevant pre-period diagnostics; the complete severity event-study coefficients remain in the replication outputs rather than the paper.

Table D2: Waiting/closure severity pre-trends—Panel A: In-person turnout

Pre-treatment election	Federal exposure	Waiting/closure severity
2014 EU	0.326	0.438^{+}
	(0.228)	(0.238)
2016 AGH	-0.228	-0.323
	(0.196)	(0.204)
2017 BTW	-0.226	-0.823^{**}
	(0.292)	(0.297)
Observations	13,500	13,330
Precinct fixed effects	Yes	Yes
Election × district fixed effects	Yes	Yes

Notes: Entries are pre-treatment coefficients relative to the 2019 European election; parentheses report precinct-clustered standard errors. The federal-exposure coefficient is in turnout percentage points per treated area, while the waiting/closure coefficient is per documented hour and uses its own, narrower estimation sample. The two columns are not effects of equivalent treatment intensities. Pre-treatment differences in the severity series limit its causal interpretation.

Table D2: Waiting/closure severity pre-trends—Panel B: Postal turnout (continued)

Pre-treatment election	Federal exposure	Waiting/closure severity
2014 EU	-0.251	-0.332^{+}
	(0.196)	(0.188)
2016 AGH	-0.231	-0.309*
	(0.149)	(0.157)
2017 BTW	-0.058	0.076
	(0.154)	(0.142)
Observations	13,500	13,330
Precinct fixed effects	Yes	Yes
Election × district fixed effects	Yes	Yes

Notes: Entries are pre-treatment coefficients relative to the 2019 European election; parentheses report precinct-clustered standard errors. The federal-exposure coefficient is in turnout percentage points per treated area, while the waiting/closure coefficient is per documented hour and uses its own, narrower estimation sample. The two columns are not effects of equivalent treatment intensities. Pre-treatment differences in the severity series limit its causal interpretation.

Table D2: Waiting/closure severity pre-trends—Panel C: Total turnout (continued)

Pre-treatment election	Federal exposure	Waiting/closure severity
2014 EU	0.074	0.106
	(0.269)	(0.277)
2016 AGH	-0.459^{+}	-0.632*
	(0.270)	(0.297)
2017 BTW	-0.284	-0.747*
	(0.301)	(0.310)
Observations	13,500	13,330
Precinct fixed effects	Yes	Yes
Election × district fixed effects	Yes	Yes

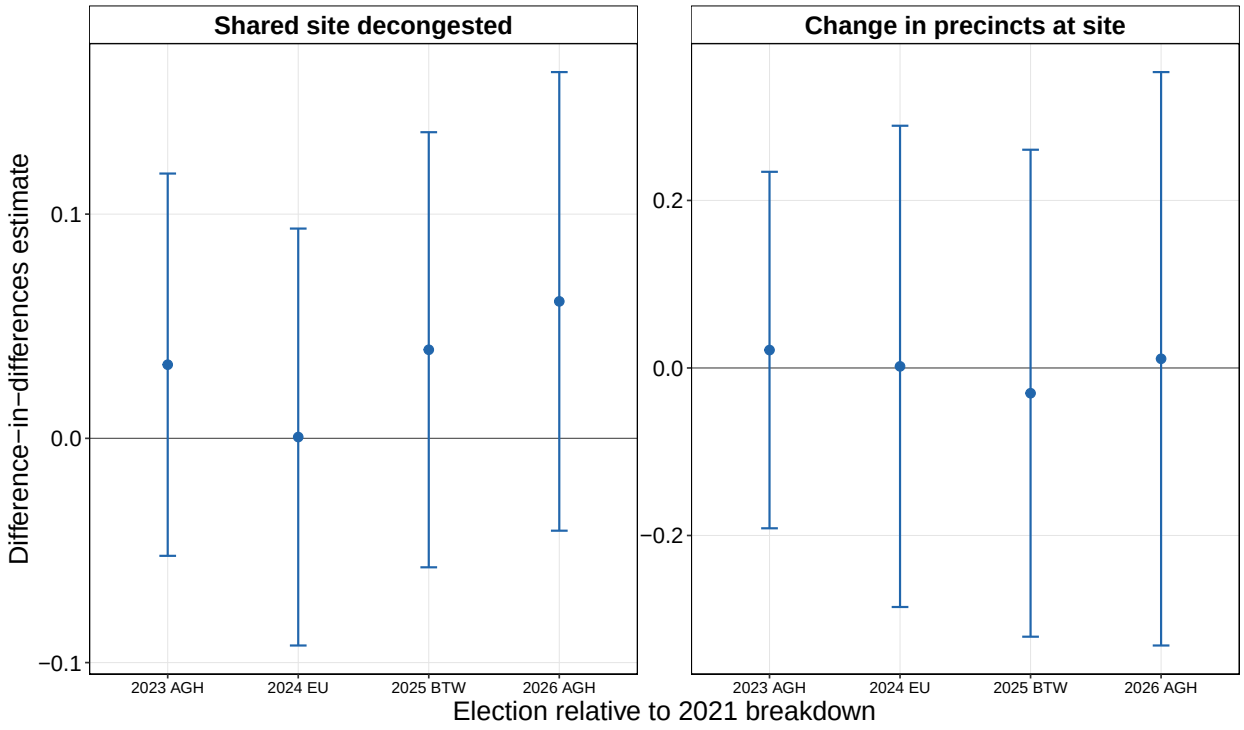
Notes: Entries are pre-treatment coefficients relative to the 2019 European election; parentheses report precinct-clustered standard errors. The federal-exposure coefficient is in turnout percentage points per treated area, while the waiting/closure coefficient is per documented hour and uses its own, narrower estimation sample. The two columns are not effects of equivalent treatment intensities. Pre-treatment differences in the severity series limit its causal interpretation.

E Physical polling-location changes

The administrative-response checks use physical polling-location points rather than treating every UWB or postal-counting location as a distinct voter-facing site. Postal-counting facilities are removed before in-person point locations are grouped into physical sites using a 50-meter clustering criterion. The analytical sample follows 2021 sites that initially hosted at least two polling-station precincts. The 50-meter rule groups nearby point records into candidate physical sites, whereas the 500-meter radius identifies alternative sites in the immediate vicinity; both are fixed operational definitions rather than thresholds validated by a separate sensitivity analysis. The outcomes describe whether original sites remain used, whether shared sites are decongested, the change in assigned precincts, and whether a different site appears within 500 meters.

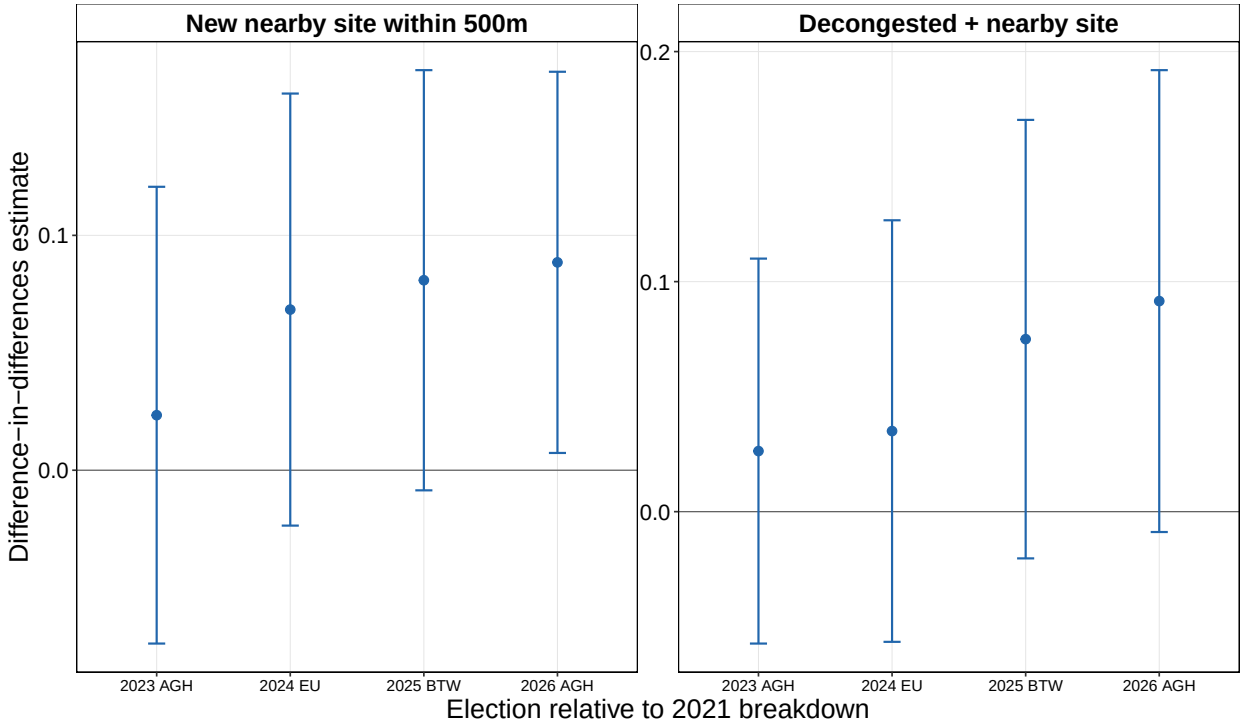
These measurements can detect differential changes in election administration after the breakdown. They are not a causal decomposition of the turnout effects: subsequent assignment decisions may respond to the earlier failure, to broader reorganization, or to other conditions affecting participation. The positive 2026 nearby-site estimate must be evaluated together with its uncertainty and the other administrative outcomes.

Figure E1: Administrative response: decongestion of shared 2021 polling locations



Notes: Unit is the 2021 physical polling location, defined by 50-meter clusters of in-person polling-location points after excluding `Briefwahl==1` postal-counting rows. The sample is restricted to locations that hosted at least two polling-station precincts in 2021. Estimates compare court-linked treated sites to clean-control sites, relative to AH21, with site fixed effects and election-by-district fixed effects. Numerical estimates for the administrative-response outcomes are reported in Table E1.

Figure E2: Administrative response: nearby new polling sites



Notes: Unit is the 2021 physical polling location, defined by 50-meter clusters of in-person polling-location points after excluding `Briefwahl==1` postal-counting rows. A nearby site is a different later in-person polling-location cluster within 500 meters of the 2021 site. The sample is restricted to locations that hosted at least two polling-station precincts in 2021. Estimates compare court-linked treated sites to clean-control sites, relative to AH21. The analysis uses the same physical-site construction as Figure E1. Numerical estimates are reported in Table E1.

Table E1: Administrative-response estimates

Outcome	Election	Estimate	Std. error	95% CI	p-value
Shared site decongested	2023 AGH	0.033	0.043	[-0.052, 0.118]	0.449
Shared site decongested	2024 European	0.001	0.047	[-0.092, 0.094]	0.990
Shared site decongested	2025 federal	0.040	0.049	[-0.058, 0.137]	0.424
Shared site decongested	2026 AGH	0.061	0.052	[-0.041, 0.163]	0.241
Change in precincts at site	2023 AGH	0.021	0.108	[-0.191, 0.234]	0.844
Change in precincts at site	2024 European	0.002	0.146	[-0.285, 0.289]	0.990
Change in precincts at site	2025 federal	-0.030	0.148	[-0.321, 0.261]	0.839
Change in precincts at site	2026 AGH	0.011	0.174	[-0.331, 0.353]	0.950
Original site still used	2023 AGH	0.004	0.021	[-0.038, 0.045]	0.868
Original site still used	2024 European	-0.002	0.032	[-0.065, 0.061]	0.952
Original site still used	2025 federal	0.006	0.029	[-0.051, 0.064]	0.833
Original site still used	2026 AGH	0.040	0.034	[-0.027, 0.107]	0.246
New nearby site within 500m	2023 AGH	0.023	0.050	[-0.074, 0.121]	0.636
New nearby site within 500m	2024 European	0.068	0.047	[-0.024, 0.160]	0.145
New nearby site within 500m	2025 federal	0.081	0.046	[-0.009, 0.170]	0.076
New nearby site within 500m	2026 AGH	0.089	0.041	[0.007, 0.170]	0.033
Decongested + nearby site	2023 AGH	0.026	0.043	[-0.057, 0.110]	0.537
Decongested + nearby site	2024 European	0.035	0.047	[-0.057, 0.127]	0.453
Decongested + nearby site	2025 federal	0.075	0.049	[-0.020, 0.170]	0.123
Decongested + nearby site	2026 AGH	0.092	0.051	[-0.009, 0.192]	0.074

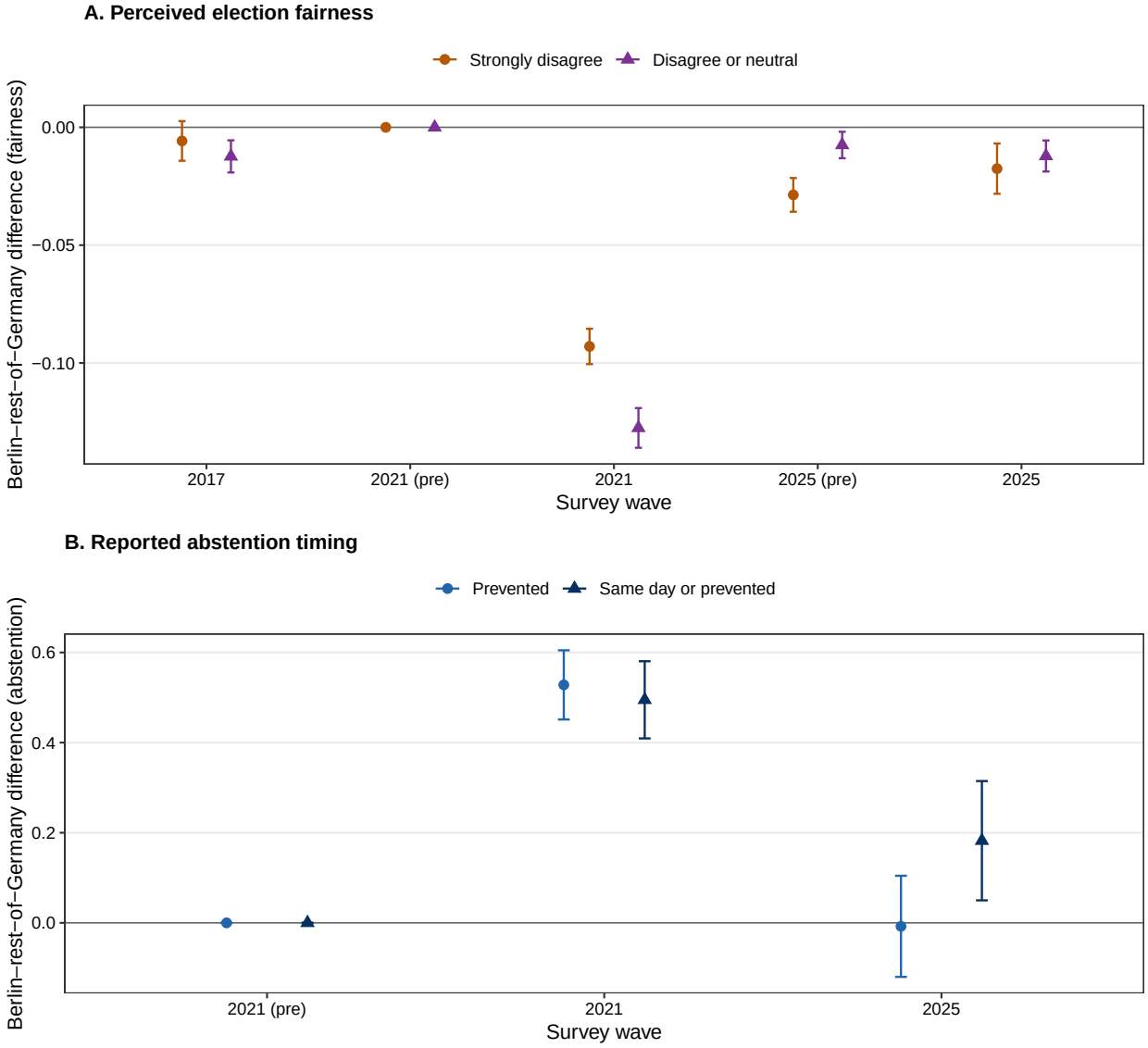
Notes: The table reports event-study estimates for the administrative-response outcomes shown in Figures E1 and E2. Unit is the 2021 physical polling location, defined by 50-meter clusters of in-person polling-location points. The sample is restricted to locations that hosted at least two polling-station precincts in 2021. Estimates compare court-linked treated sites to clean-control sites relative to AH21, with site fixed effects and election-by-district fixed effects.

F Survey context and exploratory outcomes

The German Longitudinal Election Study provides city-level context on concerns about electoral fairness and the timing of abstention decisions. Because its respondents cannot be linked to the court-documented precinct exposure used in the main event study, these comparisons establish salience rather than mechanisms of individual behavioral persistence. Figure F1 preserves the central survey evidence.

Exploratory analyses of party vote shares, neighborhoods, and age composition are not included here because they are ancillary to the participation and administrative findings. Party shares depend on the composition of participating voters, and demographic interactions remain ecological.

Figure F1: Survey evidence on perceived fairness and abstention timing



Notes: The figure reports Berlin-versus-rest-of-Germany differences in survey measures of election fairness and abstention-related responses, with panel-specific indicators and legends. The 2021 pre-election reference is normalized to zero; the scales describe distinct survey outcomes and should not be compared across panels. Because the survey lacks precinct-level exposure, these estimates are contextual salience evidence. The corresponding survey coefficients remain available in the replication outputs.

G Catchment reorganization and electorate workloads

This appendix separates two administrative outcomes: changes in the overlap of voting catchments serving fixed residential locations, and changes in the electorate size of the election-year UWB assigned to those same locations. Equation (5) defines the exact population-weighted overlap metric; Equation (6) defines the joint administrative estimation model. The first outcome measures assignment reorganization, not physical polling capacity. The second is an electorate proxy, not booths, staffing, travel time, or actual waiting, and its 2026 comparison crosses an expansion of the eligible state electorate. Source-year observations use real original precinct geometries, and the 2022 residential population is held fixed across comparisons. The 2023 state repeat inherits 2021 geometry and is not treated as an independently observed redistricting.

2026 levels and distribution

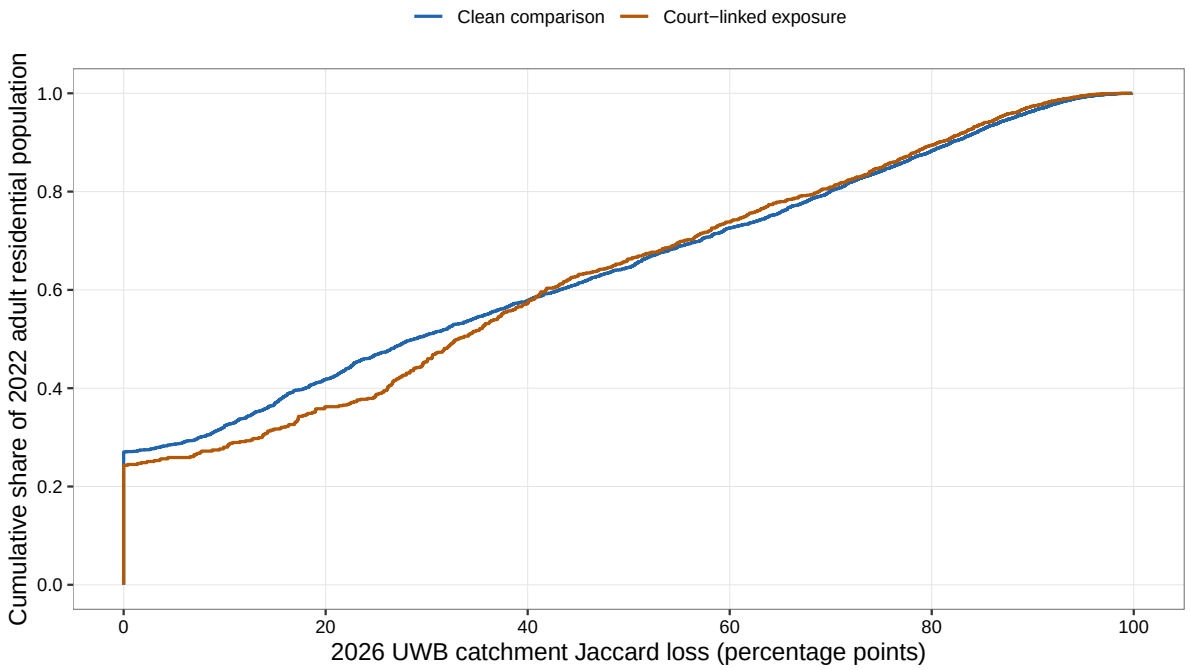
The common weighted-median split gives a descriptive high–low difference of 24.2 percentage points in court-linked areas, compared with 14.8 points in clean controls (Table G1). These unadjusted group differences illustrate the steeper size–reorganization pattern in court-linked areas; they are distinct from the adjusted one-standard-deviation interaction. Figure G1 reports the distribution across the fixed residential population.

Table G1: Population-weighted 2026 catchment reorganization by exposure and baseline electorate

Exposure	2021 electorate	Grid cells	Mean loss (pp)	Loss ≥10% (%)	Mean 2021 UWB electorate
Clean	Low	16,743	28.8	57.9	961
Clean	High	11,844	43.6	80.6	1249
Court-linked	Low	2,075	19.1	42.5	1009
Court-linked	High	5,932	43.3	83.8	1332

Notes: Mean loss and the share above 10% are population-weighted using the fixed 2022 grid of German citizens aged 18 and above. High and low baseline original-UWB electorate groups are defined by the common population-weighted median in 2021. Grid-cell counts and original-UWB workloads are descriptive, not independent precinct-level treatment effects. The full original-UWB catchments can include residential cells outside the classified 1,350-area comparison; the descriptive groups themselves contain only treated and clean-control areas.

Figure G1: Distribution of 2026 Jaccard catchment loss in the fixed residential population



Notes: Each curve shows the cumulative 2022 adult-German-citizen residential population share at or below a given 2026 population-weighted Jaccard loss. The curves are descriptive population distributions and do not condition on baseline workload or district effects.

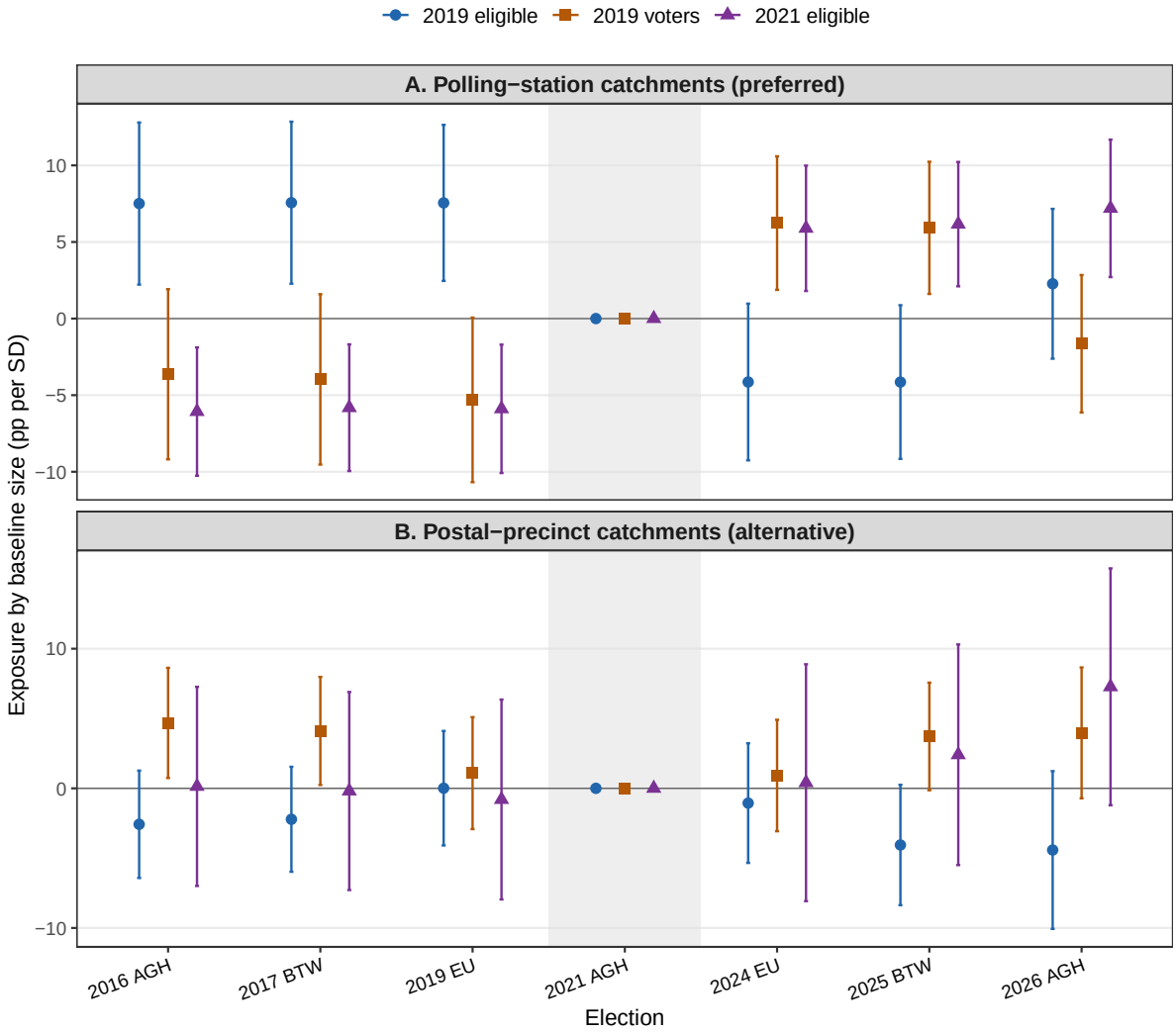
Alternative baseline measures and historical reference

The preferred regression conditions on several baseline workload measures. Figure G2 varies their roles and documents differences across election years. Any comparison of a pre-2021 catchment to 2021 is a retrospective comparison with a future administrative reference, not a placebo estimate of pre-breakdown reform. Figure G3 checks whether using a 10% threshold rather than continuous loss changes the direction of the result.

Functional form and overlap normalization

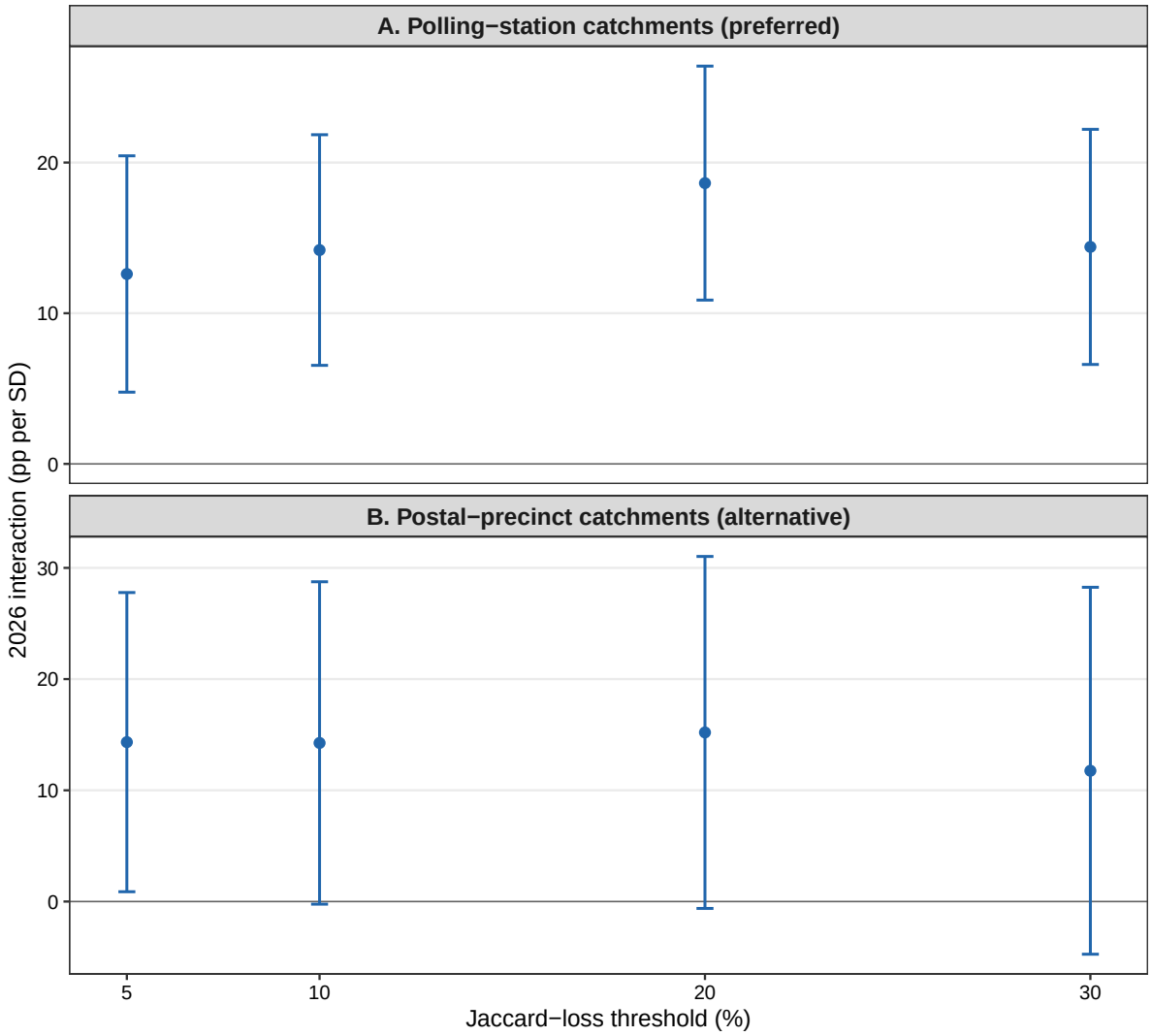
Original catchment size enters both baseline workload and the population-overlap metric, so dependence on the original size distribution is an important concern. The nonlinear model permits the exposure-specific marginal gradient to vary with original precinct size; its linear interaction component is therefore not the same constant slope contrast as in the preferred model. The alternative normalization by the original catchment population produces a similarly positive estimate.

Figure G2: Catchment reorganization across alternative baseline workload measures



Notes: Panel A (preferred) measures population-weighted changes in UWB polling-station catchments serving fixed residential grid cells. Panel B applies the analogous overlap construction to BWB postal-precinct catchments and is an alternative administrative geography, not another standard-error specification for the same outcome. Each panel compares interactions with 2019 electorate, 2019 voters, and 2021 electorate measures. The 2021 reference point is fixed at zero by construction; pre-2021 comparisons are retrospective distances to 2021 geography, not placebo estimates of administrative reform. The baseline 2021 workload is not an exogenous instrument.

Figure G3: Sensitivity of 2026 catchment reorganization to Jaccard thresholds



Notes: Panel A (preferred) defines the binary Jaccard-loss threshold for UWB polling-station catchments. Panel B repeats the construction for BWB postal-precinct catchments, a different geographical outcome. Each point is the 2026 exposure-by-baseline-size interaction in percentage points for the specified threshold, with a capped 95% confidence interval. The main-text specification instead uses continuous UWB Jaccard loss, avoiding a selected threshold.

Table G2: 2026 reorganization interaction: baseline-size functional form and overlap normalization

2026 specification	Interaction (pp)	SE: BWB21	SE: district	Grid cells
Original joint Jaccard	7.20	(2.29)	(2.14)	36,594
Quadratic 2021 size controls	4.74	(2.25)	(1.48)	36,594
Asymmetric old-catchment loss	7.93	(2.11)	(1.98)	36,594

Notes: All models use fixed 2022 residential-population weights, district fixed effects, the same four standardized 2019/2021 baseline controls, and a common 2026 classified grid-cell sample. The quadratic model adds squared standardized 2021 electorate and catchment-population terms and their exposure interactions. The alternative overlap outcome is one minus the shared population divided by the original 2021 catchment population; it is *not* the symmetric Jaccard-union denominator. These checks probe mechanical dependence on original catchment size and do not identify policy intentions. Standard errors are conventionally clustered by BWB21 area or Berlin district.

Inference for catchment reorganization

With exposure and administrative decisions geographically concentrated, inference using 12 district clusters is a necessary sensitivity to the baseline BWB21 clustering. Table G3 places conventional district uncertainty and Webb bootstrap p-values alongside the headline results.

Table G3: Catchment-reorganization interactions: conventional and Webb inference

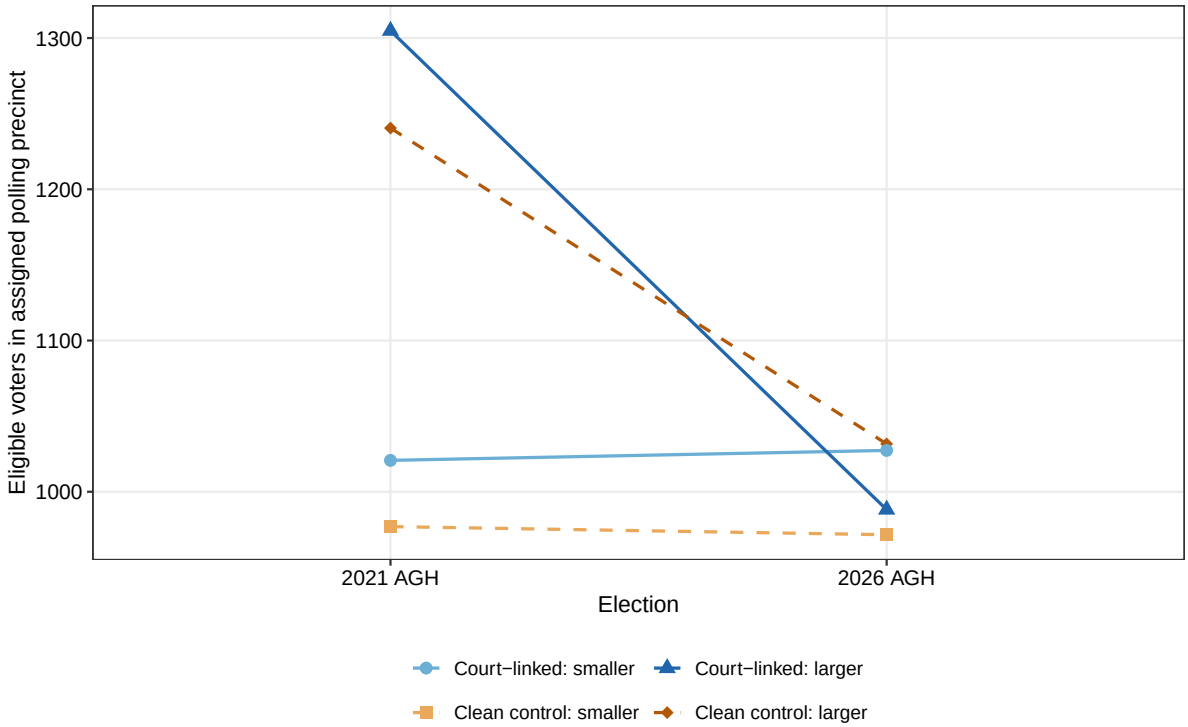
2026 outcome	Effect (pp)	SE: BWB21	SE: district	Webb p
Continuous Jaccard loss	+7.20	(2.29)	(2.14)	0.017
Jaccard loss $\geq$10%	+12.13	(3.91)	(3.35)	0.041

Notes: The reported 2026 joint-model interactions are scaled to percentage points of Jaccard loss or binary reorganization probability. Conventional standard errors are clustered at the BWB21 area or Berlin district level. Webb wild-cluster bootstrap p-values test the null of a zero 2026 exposure-by-standardized-baseline-electorate interaction using 1,999 restricted, two-sided draws clustered by Berlin district (12 clusters). They complement the conventional uncertainty estimates. Full leave-one-district-out coefficients are retained in the replication diagnostics.

Assigned UWB electorates

The final pair of exhibits reports changes in the eligible electorate assigned to the UWB serving each fixed residential location and the adjusted/joint exposure-by-baseline-size interactions. The 2021 and 2026 values are drawn from their respective election-year UWBs, then population-weighted within fixed BWB21 areas. This proxy describes assigned electorate size, with the 2026 comparison also reflecting the expanded state franchise.

Figure G4: Assigned polling-precinct electorate size by baseline exposure



Notes: The vertical axis measures eligible voters in the election-year in-person polling precinct (UWB) assigned to each fixed residential location, not an invariant original-2021 UWB boundary. Values are averaged within fixed BWB21 areas using the same 2022 residential population weights for both years. High/low groups use the population-weighted median 2021 assigned electorate; these are descriptive group means. The 2026 state franchise expansion also affects the comparison.

Table G4: Changes in assigned polling-precinct electorate size: specifications and inference

Workload model	Effect (%)	SE: BWB21	SE: district	Webb p
Basic	-1.48	(0.74)	(1.29)	—
Adjusted	-0.95	(1.05)	(1.59)	0.665
Joint	-0.78	(1.07)	(1.48)	0.689
ANCOVA	-0.95	(1.05)	(1.59)	—

Notes: The Basic, Adjusted, and Joint specifications use the 2021–2026 log change in population-weighted assigned UWB electorate size; the ANCOVA specification instead uses the log 2026 level. All are aggregated to the same 1,350 BWB21 areas. The Adjusted and ANCOVA interaction estimates and standard errors agree at the displayed precision, but are not numerically identical: including standardized $\log(1 + W_{21})$ as a covariate makes the models approximately, not exactly, equivalent because the change outcome subtracts $\log(W_{21})$. The ANCOVA row is a functional-form cross-check, not independent evidence. Baseline covariates are log-one-plus transformed and standardized at the BWB21 level using fixed population weights. Entries are exposure-by-standardized-2021-electorate interactions expressed in approximate percent units; both district- and BWB21-clustered conventional standard errors are shown, with Webb sensitivity for the adjusted and joint models. Zero-electorate residential cells are excluded from the indicated change specifications; supporting diagnostics are available in the replication materials.